What the Past Can Fix:

becoming, determination and existence in general relativity

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Abstract

Four questions about the future are often run together: whether it has become, whether it is fixed, whether it exists, and whether claims about it are already true. We separate them, with theorems where the question is mathematical and with countermodels to stated inferences where it is not. Under a locality premise, what has become for an event in any globally hyperbolic spacetime lies between its chronological and causal past, unless everything has become for everything, and a transitive local relation of becoming defined from the metric alone admits no shared present. What is fixed from an event is a different set. It contains what the causal structure guarantees, which we characterize by slices and compute in the standard models, and a particular law can fix more. In the closed dust model a massless field is fixed on part of an event's far future while its near future stays open, and every massive field is fixed exactly on the past before maximum expansion and everywhere from then on. In a smooth complete class of Einstein–scalar spacetimes, conditional on named external estimates, the whole pointed causal past and the total incoming mass leave an open ball of future central readings free, while agreement on everything outside the causal future fixes the entire development. So whether the future is open depends on which base is held fixed and on the law. Given a base and a law, the answer is the same whatever one says about whether the future exists; if the base is taken to be what exists from an event, the answer varies with the ontology. The laws give fixing no arrow of their own: for time-reversal invariant laws every such result has a mirror image. A model can have one from its global geometry, as the closed dust universe does before and after maximum expansion, but that arrow reverses with epoch, so the slogan that the past is fixed and the future open needs another source. We then ask what would be needed to reach existence. Stated bridges about constitution and persistence supply lower bounds on what is concrete from an event. A restricted ontology needs an upper bound, an unrestricted one a stronger lower bound, and neither comes from the results above. We sort the stated candidates, and record which inferences fail. No temporal ontology is selected.

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1 Introduction

Is the future real? Is it already fixed? Is what we say about it already true? Has it become? They are often treated as one question. An open future is one that does not yet exist, whose outcome nothing settles, about which there is no fact yet. Philosophers of time have long known that the pieces can be pulled apart in principle. A growing block in Broad's sense (Broad, 1923), with deterministic laws added, has an unreal future that is settled. A later-grounded bivalent theory has truths about what does not yet exist (Correia and Rosenkranz, 2018). What has been missing is an account, in general relativity, of how far apart the pieces actually lie, and of exactly what fixes what.

This paper gives that account. It has one thesis:

> What has become for an event, what is fixed from it, what exists from it and what is true from it are four different relations. General relativity constrains the first sharply under a locality premise. It makes the second a mathematical question, once a law and a comparison are named, and we answer it exactly in the models treated. Nothing here determines the third or the fourth, and the first two do not determine them.

The thesis is defended by results of four kinds. We state them here with their conditions.

Becoming (Part I). Stein's theorem, with its contextual conditions, says that in Minkowski spacetime a transitive, symmetry-invariant relation of becoming that contains the causal past and leaves something unsettled is the causal past (Stein, 1968, 1991). Clifton and Hogarth give the form whose alternatives are the chronological past, the causal past and the universal relation (Clifton and Hogarth, 1995). A curved spacetime can have no symmetries, and the single-spacetime statement is then false. Replacing symmetry by the local covariance of quantum field theory restores the result: either everything has become for everything in every globally hyperbolic spacetime, or in every such spacetime becoming lies between the chronological and causal past (Theorem 8). The residual freedom is the light cone, and it can depend on curvature. For transitive relations that contain the chronological past and are defined from the metric alone, locality, a shared present and non-trivial becoming cannot all be had (Corollary 9). Without transitivity they can: the bow tie of Section 9 has all three. In flat spacetime that is the only local way: a local symmetric relation among causally unrelated events is empty or is spacelike separation itself (Corollary 12). A rival with a preferred clock changes the category and is not excluded (Section 3.7).

Determination on fixed backgrounds (Part II). What the past fixes is a different set. Theorem 24 characterizes what the causal structure guarantees, by a test on slices. Theorem 37 then lays out what can happen. The fixed set can equal the causal past, exceed what is guaranteed on an isolated part of the far future, or be everything; the same metric gives different answers under different laws; and an exact verdict can flip while the hidden fields change by $O(m^2)$ in a stated norm (equation (3)). In one closed universe a massless field is fixed at the end of an event's worldline before the middle (Theorem 33), and every massive field has no such islands (Theorem 35).

Determination with gravity (Part III). Backreaction does not close the gap. In a smooth, complete, inextendible class of Einstein–scalar spacetimes, the entire pointed causal past of an event and the total incoming mass leave an open ball of future central scalar readings free (Theorem 38), yet agreement on the open region outside the event's causal future fixes the whole development (Theorem 41). Both statements concern the same class. The first is conditional on external estimates of Luk, Oh and Yang. Its moral is already present for the linear wave equation; what coupling adds is that the moral survives gravity, a fixed mass and the axis. Which base is compared therefore matters more than any general slogan about openness.

Existence and truth (Part IV). None of this reaches existence. We show what would: a stated principle bounding what is concrete from an event. Stated bridges from the physics give lower bounds. A restricted ontology needs an upper bound, and an unrestricted one needs a stronger lower bound (Section 14). We sort the stated candidates. The useful result is a diagnostic for these candidates: an index on "concrete from b" is either defined, and then trivial and shared, or primitive, and then a thesis the event realist rejects. Three candidates need support of a different kind. Truth policies are independent again (Section 12). Fixing has no arrow of its own, so the past as a base needs a separate account (Section 13). Section 16 records which inferences fail.

What is not claimed. No temporal ontology is selected. The results are conditional as stated, and Appendix C gives the status of each. Parts of this material overlap prior work, which is credited where it is known; a claim of priority would need an audit this paper has not completed.

2 Setting

Spacetimes are smooth, connected, four-dimensional, oriented and time-oriented. Unless stated otherwise they are globally hyperbolic. We follow Clifton and Hogarth in taking I^-, J^- to be reflexive (Clifton and Hogarth, 1995, p. 360). We write $I^-_{\mathrm{s}}(p)$ for the strict chronological past, so $I^- = I^-_{\mathrm{s}} \cup \{p\}$, $x \ll y$ for $x \in I^-_{\mathrm{s}}(y)$, and $x \leq y$ for $x \in J^-(y)$. Standard causal facts are cited from Minguzzi and Sánchez (Minguzzi and Sánchez, 2008), abbreviated MS, whose $I^\pm$ are our strict sets. A *Cauchy time function* is a continuous surjection $t : M \to (\alpha, \beta)$ strictly increasing along future causal curves, with Cauchy level sets $S_c = t^{-1}(c)$ (Bernal and Sánchez, 2005).

Fix an event b, which we call the standpoint. We attach to it the following.

Object	Meaning	Decided by
$B_M(b)$	what has become for b	a locality premise (Part I)
$\mathrm{Det}(b)$	what the causal structure guarantees from $I^-_{\mathrm{s}}(b)$, fixed by every law with the domain-of-dependence property	causal structure (Part II)
$F_{\mathcal{L}}(S)$	$x \in F_{\mathcal{L}}(S)$ if any two solutions agreeing on S agree at x; $F_{\mathcal{L}}(b) := F_{\mathcal{L}}(I^-_{\mathrm{s}}(b))$	law, class and base (Parts II, III)
E_b	what is concrete from b	a bridge premise (Part IV)
truth at b	which claims about later events hold from b	a semantic policy (Part IV)

These are not five names for one thing, and the paper is about the gaps between them. Three points of method follow from keeping them apart.

First, name the comparison. Agreement of accessible records, agreement of an intrinsic causal past, and agreement of an entire proposed realm are three different contracts. A failure under one

is not a failure under another. Here we mostly compare exact agreement of classical fields on a named region.

Second, separate physical premises, definitions and bridges. A definition that already assumes an ontology cannot establish it. We mark bridge premises as such.

Third, keep models in their domains. A finite example or a closed dust universe shows what is possible under stated hypotheses. Neither is a claim about the actual cosmology.

All four relations are defined on a classical spacetime of events. A theory without events would remove their domain and not refine them, and we do not ask which of them has a successor there.

Part I. Becoming

3 Becoming under locality

Definition 1. GH is the category whose objects are globally hyperbolic spacetimes and whose morphisms $\psi : M \to M'$ are smooth isometric embeddings preserving orientation and time orientation, with $\psi(M)$ open and causally convex in M'. This is the category Loc of locally covariant quantum field theory (Brunetti et al., 2003, §2), (Fewster and Verch, 2012, §2.2).

Definition 2. A *becoming theory* assigns to each $M \in \mathsf{GH}$ a relation $B_M \subseteq M \times M$ (read $x\, B_M\, b$ as "x has become for b") such that, for every M:

(LC) for every morphism $\psi : M \to M'$, $x\, B_M\, b \iff \psi(x)\, B_{M'}\, \psi(b)$;

(T) B_M is transitive;

(CB) $I^-_M \subseteq B_M$.

(CB) is Clifton and Hogarth's chronological becoming; they derive it from "worldline becoming" and "worldline transitivity" (Clifton and Hogarth, 1995, Lemma, p. 364), and the derivation carries over verbatim to curved spacetime. (LC) contains invariance under every orientation- and time-orientation-preserving isometry of each M, which on Minkowski spacetime $\mathbb{M}$ is the proper orthochronous Poincaré group. Its further content is *locality*: if $N \subseteq M$ is open, causally convex and contains x and b, then whether x has become for b is settled by N alone.

Remark 3 (What (LC) contains)**.** Three premises should be kept apart. *Naturality*: one rule for all objects, preserved by every morphism. *Reflection*: the converse direction of (LC), by which every causally convex region containing x and b settles the verdict; the proof of Theorem 8 uses it along $N \hookrightarrow M$ and $N'' \hookrightarrow M'$ in part (a), and along $\psi|_N : N \to M'$ and $N'' \hookrightarrow \mathbb{M}$ in part (b), and functoriality in quantum field theory supplies no such direction. *Off-shell universality*: Remark 4. "Local" here is geometric: independence of what lies around a region, the sense of Brunetti et al. (2003, §1). It is not no-signalling or microcausality, and neither of those implies it. The case for (LC) is that a relation defined from geometry alone should not depend on what lies outside a causally convex region containing the related events. That is a premise about becoming, not a consequence of relativity. Reflection is strong on its own. Before transitivity or (CB) is used, it excludes every relation that depends on the separation of spacelike events (Proposition 10). For example, the relation "x and b lie in a common diamond $I^+(z_1) \cap I^-(z_2)$ with $d(z_1, z_2) < \ell_0$" is defined from the metric in every spacetime and violates Reflection, since a thin causally convex tube from x to b need not contain z_1 or z_2.

Remark 4 (The off-shell premise)**.** GH contains every globally hyperbolic metric, not only solutions of Einstein's equations; the category is defined kinematically in (Brunetti et al., 2003; Fewster and Verch, 2012). Theorem 8 uses this: its proof builds auxiliary spacetimes that need not satisfy any field equation with reasonable matter. This is natural for a kinematic notion of becoming, defined (as Stein and Clifton–Hogarth define theirs) from geometry alone, but it is a premise. Section 3.6 examines what happens without it. On some natural classes of spacetimes the conclusion then fails (Proposition 20).

3.1 Why one spacetime is not enough

Proposition 5. *Let M be globally hyperbolic and $x, b \in M$ spacelike-separated. Then $R = J^- \cup (J^-(x) \times J^+(b))$, where (e, z) in the second set means e has become for z, is reflexive and transitive, contains I^-, relates x to b, and is neither I^-, J^- nor $\mathcal{U}$. If M has no nontrivial isometries, R is invariant under all of them.*

Proof. Write $P = J^-(x) \times J^+(b)$. Compositions $J^- \circ J^-$, $J^- \circ P$ and $P \circ J^-$ land in J^- or P because $\leq$ is transitive. A composition $P \circ P$ would need some q with $b \leq q \leq x$, which is impossible. The pair (b, x) is not in R, so $R \neq \mathcal{U}$. □

So for a single generic spacetime, Stein-type uniqueness fails completely. The next example shows that uniformity across spacetimes is still not enough without locality.

Proposition 6. *Let $\tau_M(p)$ be the supremum of the lengths of past-directed causal curves from p (the cosmological time of Andersson, Galloway and Howard (Andersson et al., 1998)). Put $B_M = \{(x, b) : \tau_M(x) \leq \tau_M(b)\}$ if τ_M is finite on M, and $B_M = J^-_M$ otherwise. Then B is invariant under all isometries, each B_M is reflexive and transitive and satisfies (CB), and B relates spacelike pairs in every spacetime where τ is finite (e.g. big-bang Friedmann–Lemaître–Robertson–Walker models). It is not a becoming theory: it violates (LC).*

Proof. Isometries preserve lengths of causal curves. If $x \ll b$, concatenating past curves from x with a timelike curve from x to b gives $\tau(b) \geq \tau(x) + d(x, b) > \tau(x)$, so (CB) holds; transitivity is that of $\leq$ on the reals. Failure of (LC): the Milne wedge $W = I^+_{\mathbb{M}}(0)$ is open, causally convex and globally hyperbolic, and its inclusion into $\mathbb{M}$ is a morphism. On W, $\tau_W(p)$ is the proper time from the origin, finite, so B_W relates distinct points of the same hyperboloid $\{\tau_W = c\}$, which are spacelike-separated; on $\mathbb{M}$, $\tau = \infty$ and $B_{\mathbb{M}} = J^-$ does not relate them. □

3.2 The classification

Theorem 7 (Clifton–Hogarth)**.** *For every becoming theory, $B_{\mathbb{M}} \in \{I^-, J^-, \mathcal{U}\}$.*

Proof. Proper orthochronous Poincaré maps are automorphisms of $\mathbb{M}$ in GH, so $B_{\mathbb{M}}$ is invariant under them by (LC). Clifton and Hogarth's Theorem 1 (Clifton and Hogarth, 1995, p. 365) gives the conclusion from this invariance, transitivity and chronological becoming. Its proof (pp. 365–367) uses only point-preserving spatial rotations, boosts and translations, and they note that it does not use spatial reflections (pp. 385–386). Their stated hypothesis is invariance under all time-orientation-preserving isometries, reflections included; using only the proper orthochronous subgroup is a weakening whose adequacy rests on their proof, not their statement. □

Theorem 8 (No becoming outside the causal past)**.** *Let B be a becoming theory.*

(a) If $B_{\mathbb{M}} \neq \mathcal{U}$, then $B_M \subseteq J^-_M$ for every $M \in \mathsf{GH}$.

(b) If $B_{\mathbb{M}} = \mathcal{U}$, then $B_M = \mathcal{U}_M$ for every $M \in \mathsf{GH}$.

Hence either B is universal everywhere, or $I^-_M \subseteq B_M \subseteq J^-_M$ for every M.

Call B *universal* if $B_M = \mathcal{U}_M$ for every M. By Theorem 8 this holds if and only if $B_{\mathbb{M}} = \mathcal{U}$, and if $B_M = \mathcal{U}_M$ for one M then it holds for all; otherwise B is *non-universal*.

Corollary 9 (No shared present)**.** *Call x and b* co-present *in M if $x\, B_M\, b$ and $b\, B_M\, x$. For every non-universal becoming theory, co-presence is the identity relation in every $M \in \mathsf{GH}$. Equivalently, no becoming theory is local, non-trivial and admits two distinct co-present events in some globally hyperbolic spacetime.*

Proof. By Theorem 8(a), $x \le b$ and $b \le x$, and globally hyperbolic spacetimes are causal, so $x = b$. □

The next three statements use (LC) alone. Call an assignment $M \mapsto R_M \subseteq M \times M$ satisfying (LC) a *local relation*; no transitivity and no (CB) is assumed. Write $d_M(x, b)$ for the supremum of the lengths of future causal curves from x to b.

Proposition 10 (No spacelike scale)**.** *For every local relation R, $R_{\mathbb{M}}$ contains no pair of spacelike-separated events or every such pair.*

Proof. By (LC), $R_{\mathbb{M}}$ is invariant under the proper orthochronous Poincaré group, which is transitive on ordered spacelike pairs of a given interval. So there is $f : (0, \infty) \to \{0, 1\}$ with $x\, R_{\mathbb{M}}\, b \iff f(s) = 1$, s the interval.

Let $C_\ell = \mathbb{R}_t \times \mathbb{R}^2 \times S^1_\ell$ be the flat cylinder with circle of length ℓ; it lies in GH, and $\{t = 0\}$ is a Cauchy hypersurface. Fix $0 < a < \ell$ and $0 < 2\rho < \min\{a, \ell - a\}$. Let $x = (0, 0, 0)$, $b = (0, 0, a)$ in coordinates (t, y, θ), and

$$S_1 = \{t = 0,\ |y| < \rho,\ \theta \in (-\rho, a + \rho)\}, \qquad S_2 = \{t = 0,\ |y| < \rho,\ \theta \in (a - \rho, \ell + \rho)\}.$$

Each S_i is acausal and contains x and b. Put $N_i = \operatorname{int} D(S_i)$. Let $\pi : \mathbb{M} \to C_\ell$ be the covering map and $\tilde{S}_i$ a lift of S_i, a straight tube in $\{t = 0\}$. We claim π maps $\operatorname{int} D_{\mathbb{M}}(\tilde{S}_i)$ isometrically onto N_i. Inextendible causal curves lift, so $\pi(D_{\mathbb{M}}(\tilde{S}_i)) \subseteq D(S_i)$. Conversely let $p \in D^+(S_i)$ at height τ over $q \in S_i$, and let $\tilde{q} \in \tilde{S}_i$ lift q. The straight past causal rays from p meet $\{t = 0\}$ once, in $\pi(\bar{B}(\tilde{q}, \tau))$, which therefore lies in S_i. The preimage of S_i is a disjoint union of translates of $\tilde{S}_i$, and the ball is connected and contains $\tilde{q}$, so $\bar{B}(\tilde{q}, \tau) \subseteq \tilde{S}_i$ and $(\tau, \tilde{q}) \in D^+_{\mathbb{M}}(\tilde{S}_i)$ projects to p. The case of D^- is the mirror image, and π is injective over $\tilde{S}_i$.

N_i is causally convex: if $p, q \in D(S_i)$, $p \le r \le q$ and λ is an inextendible causal curve through r, then the curve back from q to r followed by the past half of λ meets S_i, and so does the curve from p to r followed by the future half of λ; if neither meeting point is on λ there are $s_2 \le r \le s_1$ in S_i, and acausality gives $s_1 = s_2 = r$. Interiors follow by moving p, q slightly. N_i is connected, and it is globally hyperbolic because it is isometric to $\operatorname{int} D_{\mathbb{M}}(\tilde{S}_i)$, whose causal diamonds are compact diamonds of $\mathbb{M}$. So $N_i \hookrightarrow C_\ell$ and $N_i \cong \operatorname{int} D_{\mathbb{M}}(\tilde{S}_i) \hookrightarrow \mathbb{M}$ are morphisms. In the first lift the endpoints are at interval a, in the second at interval $\ell - a$. Applying (LC) four times,

$$f(a) = [x\, R_{N_1}\, b] = [x\, R_{C_\ell}\, b] = [x\, R_{N_2}\, b] = f(\ell - a).$$

Given $s, s' > 0$ take $a = s$, $\ell = s + s'$. So f is constant. □

The proof uses only flat spacetimes, so it holds on any full subcategory of GH that contains $\mathbb{M}$ and the cylinders and is closed under restriction to causally convex open subsets, including the vacuum and dominant-energy subcategories.

Corollary 11 (Local relations on Minkowski spacetime)**.** *For a local relation R, $R_{\mathbb{M}}$ is a union of: the diagonal or nothing; all future lightlike pairs or none; all past lightlike pairs or none; all spacelike pairs or none; and the future (respectively past) timelike pairs whose proper time lies in an arbitrary set T_+ (respectively T_-) $\subseteq (0,\infty)$. Every such union is the restriction to $\mathbb{M}$ of a local relation.*

Proof. $R_{\mathbb{M}}$ is a union of orbits of the proper orthochronous Poincaré group, and Proposition 10 treats the spacelike orbits. Conversely, equality, the horismos relations, the absence of any causal relation, and the conditions $d_M(x,b) \in T_+$, $d_M(b,x) \in T_-$ are preserved and reflected by morphisms, because causal curves between points of a causally convex set stay in it. □

Corollary 12 (Local presents)**.** *Let R be a symmetric local relation that relates no two distinct causally related events. Then for each $b \in \mathbb{M}$ the set of events R-related to b is contained in $\{b\}$, or is the set of all events spacelike to b, possibly together with b.*

So a proper-time threshold survives locality and a spacelike one does not. The causal hull of a timelike pair is the diamond between them, which every causally convex region containing the pair contains and which fixes their proper time. The causal hull of a spacelike pair is the two points.

Corollary 13 (Worldline-dependent becoming)**.** *Suppose that for each $M \in \mathsf{GH}$, each inextendible timelike curve λ in M and each $b \in \lambda$ a set $B_{M,\lambda}(b)$ is given, satisfying Clifton–Hogarth's worldline becoming ($I^-(b) \cap \lambda \subseteq B_{M,\lambda}(b)$) and worldline transitivity ($a \in B_{M,\lambda}(b)$ and $b \in B_{M,\lambda'}(c)$ imply $a \in B_{M,\lambda'}(c)$) (Clifton and Hogarth, 1995, pp. 362–363), and local covariance in the form: for a morphism $\psi : N \to M$, an inextendible timelike λ_N in N and any inextendible timelike λ in M extending $\psi \circ \lambda_N$, $a \in B_{N,\lambda_N}(b) \iff \psi(a) \in B_{M,\lambda}(\psi(b))$. Then $B_{M,\lambda}(b)$ does not depend on λ, and the resulting relation is a becoming theory. Hence Theorem 8 applies to worldline-dependent becoming.*

Clifton and Hogarth anticipated that extending their Theorem 2 would be harder still ("much less to extend Theorem 2", p. 385). Under local covariance it is not: as they themselves note (p. 363), worldline transitivity makes observers whose worldlines cross agree, and the remaining work is done by Theorem 8.

Write $E^- = J^- \setminus I^-$ for the horismos (lightlike-only relation).

Theorem 14 (Classification of the residual freedom)**.** *Let B be a non-universal becoming theory and $N_M = B_M \cap E^-_M$. Then N is locally covariant and satisfies*

(NC) if $(x,y),(y,b) \in N_M$ and $(x,b) \in E^-_M$, then $(x,b) \in N_M$.

Conversely, if $N \subseteq E^-$ is locally covariant and satisfies (NC), then $B = I^- \cup N$ is a becoming theory. On $\mathbb{M}$, N is empty or all of E^-.

Theorem 15 (Non-uniqueness)**.** *Let $N_M(x,b)$ hold iff $x \in E^-(b)$, $x \neq b$, and some lightlike geodesic segment from x to b contains a point at which the Weyl tensor is nonzero. Then $I^- \cup N$ is a becoming theory equal to I^- on $\mathbb{M}$, equal to J^- on any spacetime with nowhere-vanishing Weyl tensor (such as the Schwarzschild exterior), and strictly between I^- and J^- on spacetimes that are flat on a nonempty causally convex open set and have nonzero Weyl tensor somewhere. It depends only on the conformal structure (not merely the curvature: conformally flat spacetimes such as FLRW give I^-).*

Proposition 16 (Regularity)**.** *Let B be non-universal. If every B_M is closed in $M \times M$, then $B = J^-$. If every $B_M \setminus \Delta_M$ is open, then $B = I^-$.*

Proofs are in Appendix A.

3.3 Discrete and extreme cases

3.4 Causal sets

Let CS have locally finite posets as objects and order-embeddings with order-convex image as morphisms. A becoming theory on CS is defined as before, with (CB) read as $x \leq b \Rightarrow x\,B\,b$.

Proposition 17. *Every becoming theory on* CS *is either the causal order in every causal set or the universal relation in every causal set.*

Proof. If x, b are incomparable, $\{x, b\}$ is convex, so both embeddings of the two-element antichain are morphisms: the verdict on incomparable pairs is a single symmetric bit β, the same in every causal set. If $x > b$, the interval $I = [b, x]$ is finite and convex, so $x\,B\,b$ depends only on I. If $\beta = 1$, in $I \sqcup \{z\}$ with z incomparable to all, $x\,B\,z\,B\,b$; so every such verdict is positive and B is universal. If $\beta = 0$ and $x\,B\,b$ for some $x > b$, adjoin e with $e < x$ only: I remains convex, so $x\,B\,b$ there, $e\,B\,x$ by (CB), and $e\,B\,b$ with e incomparable to b, contradicting $\beta = 0$. □

Wüthrich and Callender state that "The analogue of this theorem, however, is straightforwardly false in CST" (Wüthrich and Callender, 2017, §3). Their counterexample, on the diamond causal set $d < b, c < a$, adds the incomparable pair $\{b, c\}$ to the causal order and is invariant under that causal set's automorphisms. It is not the restriction of any becoming theory in the present sense: relating b to c fixes $\beta = 1$, and Proposition 17 then relates b to d, which their relation excludes. Their own footnote 5 observes that the counterexample "could be a small proper part of a much larger causet"; (LC) is exactly the requirement that the relation on such a part agree with the relation in the larger causal set. The diagnosis matches the continuum case: invariance within one structure is too weak, and locality across structures restores Stein's theorem, here with no residual freedom because a causal set has no light-cone boundary. Their claim is true on its own terms; the present result uses a stronger, cross-structure hypothesis. Note also that CS includes posets that are not manifoldlike (antichains, disjoint unions), and the proof uses them: this is the causal-set counterpart of the off-shell premise of Remark 4.

3.5 Gödel universes are isolated

Proposition 18. *Extend* GH *to all oriented, time-oriented spacetimes with the same morphisms. If M is totally vicious ($p \in I^-_{\mathrm{s}}(p)$ for all p), then every morphism into M is onto, every theory satisfying (CB) has $B_M = \mathcal{U}_M$, and there is no morphism between M and any causal spacetime, in either direction.*

Proof. In a totally vicious spacetime $I^-(p) = I^+(p) = M$ for every p (MS §3.1). Let V be nonempty, open and causally convex, $p \in V$ and $q \in M$. There are timelike curves $p \to q$ and $q \to p$; their concatenation is a causal curve with both endpoints in V, so it lies in V, and $q \in V$. Thus $V = M$ (compare MS Remark 2.12). (CB) gives $B_M \supseteq I^-_M = M \times M$. A morphism $N \to M$ from a causal N would be onto, hence an isometry, which is impossible since M has closed timelike curves; a morphism $M \to N$ would map a closed timelike curve of M to one of N. □

Gödel's universe is totally vicious, so universal becoming there is forced; this is Gödel's first step, in Stein's terms. But local covariance transmits constraints only along morphisms, and by Proposition 18 no morphism connects a totally vicious spacetime with a globally hyperbolic one. Gödel's modal step, from what holds in such a universe to what holds in ours (Gödel, 1949, pp. 561–562), as quoted in (Monton, 2006, §6), requires that becoming be the same feature across nomologically possible universes. Local covariance is one precise way of requiring a rule to be the same across universes, and it transmits nothing out of totally vicious ones. A version of Gödel's step that does go through would need a different cross-universe link. Proposition 6 shows what such links look like: natural but non-local. Compare the conceptual responses of Savitt (2005), Dieks (2006) and Aames (2022), which locate becoming locally along worldlines.

3.6 Energy conditions

Let $\mathsf{GH}_{\mathrm{DEC}}$ be the full subcategory of spacetimes whose Einstein tensor satisfies the dominant energy condition. It is closed under restriction to causally convex open subsets and contains $\mathbb{M}$.

Proposition 19 (The auxiliary spacetime is unavailable)**.** *If $M \in \mathsf{GH}_{\mathrm{DEC}}$ is flat on a neighbourhood of some totally geodesic Cauchy surface, then M is flat. In particular the spacetime M' of Appendix A lies in $\mathsf{GH}_{\mathrm{DEC}}$ only if g is flat on T_{2r}.*

Proof. On the Cauchy surface S the metric is flat and the Einstein tensor vanishes. By the conservation theorem for matter obeying the dominant energy condition (Hawking and Ellis, 1973, §4.3), the Einstein tensor vanishes on its domain of dependence, i.e. everywhere. In the form we use: let t be a Cauchy temporal function with $S = t^{-1}(0)$ (Bernal and Sánchez, 2006, Thm. 1.2) and $p \in J^+(S)$. The set $J^-(p) \cap J^+(S)$ is compact, and it is the union of the segments of integral curves of ∇t that run from S to $\partial J^-(p)$ (a future-directed curve that leaves the past set $J^-(p)$ does not return). Minguzzi's form of the theorem (Minguzzi, 2015, Thm. 1), which needs no time function, then gives $G = 0$ at p from $G = 0$ on S; $J^-(S)$ is the time reverse. For M', any slice $\{t = c\}$ with $c < -3r$ is totally geodesic and has a flat neighbourhood. A vacuum globally hyperbolic spacetime with flat data $(\Sigma, h, 0)$ on a Cauchy surface is a development of those data and so embeds isometrically in their maximal development (Choquet-Bruhat and Geroch, 1969, Thm. 3). That maximal development is the flat static spacetime $\mathbb{R} \times \Sigma$, $-dt^2 + h$ (not Minkowski in general; e.g. $\Sigma = T^3$): it is a development, and if a development properly extended it, a point p outside its image would lie on a timelike geodesic from Σ whose image under the extension coincides with the corresponding complete geodesic of $\mathbb{R} \times \Sigma$, so p would lie in the image. Hence the spacetime is flat. □

The same reasoning constrains any repair that keeps the original spacetime and inserts small flat pockets near the points w, v and the detour of the proof of Theorem 8: in a vacuum region whose causal past on some Cauchy surface is vacuum and unmodified, the conservation theorem forces the modified spacetime to be vacuum there with the same data, so no such pocket can be inserted there. Pockets must be fed by matter brought in from the past and routed around $D(\Sigma)$ afterwards. The proof of Theorem 8 goes through verbatim in $\mathsf{GH}_{\mathrm{DEC}}$ given the following sufficient condition, which we leave open.

> **Pocket problem.** Given $M \in \mathsf{GH}_{\mathrm{DEC}}$ and Σ as in Lemma 51, find $M'' \in \mathsf{GH}_{\mathrm{DEC}}$ containing an isometric copy of a neighbourhood of $D_M(\Sigma)$ whose image is a domain of dependence, and exactly flat causally convex regions containing points $w \ll x$ and $v \gg b$ joined by the flat domain of dependence of a slab.

The constraint is an inequality, not a field equation, so constructions can be kinematic, and explicit models with flat regions chronologically adjacent to curved dominant-energy regions exist (Vaidya spacetimes before an incoming, or after an outgoing, null-dust shell). Thin shells will not suffice in general, since null junctions require matching intrinsic geometry. A solution would give Theorem 8 for spacetimes with physically reasonable matter; a vacuum version would require bounded interior flat regions adjacent to arbitrary curved vacuum regions, which we have not found in the gluing literature (an abstract-level survey: Carlotto–Schoen, Corvino–Schoen, Aretakis–Czimek–Rodnianski, Czimek–Rodnianski).

A class on which the theorem fails. The off-shell premise cannot be dropped in general. Let $\mathsf{GH}_{\mathrm{Scal}}$ be the full subcategory of spacetimes whose Ricci scalar Scal is non-increasing along every future-directed causal curve. It is closed under restriction to causally convex open subsets, and contains $\mathbb{M}$, every vacuum spacetime, and the spatially flat expanding dust Friedmann–Lemaître–Robertson–Walker models, where Scal is a positive multiple of the density.

Proposition 20 (A curvature clock)**.** *On* $\mathsf{GH}_{\mathrm{Scal}}$ *each of the following satisfies (LC), is transitive and contains* I^-*:*

(i) $J^-_M \cup \{(x,b) : \mathrm{Scal}(x) > \mathrm{Scal}(b)\}$*;*

(ii) $J^-_M \cup \{(x,b) : \mathrm{Scal}(x) \geq \mathrm{Scal}(b),\ \mathrm{Scal}(x) > 0\}$*;*

(iii) $\{(x,b) : \mathrm{Scal}(x) \geq \mathrm{Scal}(b)\}$.

On $\mathbb{M}$*, (i) and (ii) are* J^- *and (iii) is* $\mathcal{U}$*. On expanding dust models (i) relates every event to every event of lower density, (ii) and (iii) are the cosmic-time preorder, and the constant-density hypersurfaces are shared presents for (ii). Hence Theorem 8(a) and (b) and Corollary 9 all fail with* $\mathsf{GH}_{\mathrm{Scal}}$ *in place of* GH*.*

Proof. Morphisms preserve Scal and preserve and reflect J^-, which gives (LC). In $\mathsf{GH}_{\mathrm{Scal}}$, $x \leq y$ implies $\mathrm{Scal}(x) \geq \mathrm{Scal}(y)$, which gives (CB) and the mixed cases of transitivity. For (ii): if $x \leq y$, $\mathrm{Scal}(y) \geq \mathrm{Scal}(b)$ and $\mathrm{Scal}(y) > 0$, then $\mathrm{Scal}(x) \geq \mathrm{Scal}(y) > 0$ and $\mathrm{Scal}(x) \geq \mathrm{Scal}(b)$; if $\mathrm{Scal}(x) \geq \mathrm{Scal}(y)$, $\mathrm{Scal}(x) > 0$ and $y \leq b$, then $\mathrm{Scal}(x) \geq \mathrm{Scal}(y) \geq \mathrm{Scal}(b)$. The remaining claims are read off. □

None of these is a becoming theory on GH, where transitivity fails. The third is the rule $\mathrm{Scal}(x) \geq \mathrm{Scal}(b)$ that locality alone does not exclude; what excludes it on GH is that (T) and (CB) are demanded on every globally hyperbolic geometry. The Milne wedge of Proposition 6 is no objection to it: the wedge is flat, and the rule returns J^- there and on $\mathbb{M}$ alike. $\mathsf{GH}_{\mathrm{Scal}}$ is not a realistic class (a star has a spacelike density gradient), and which class a becoming rule must cover is a premise we do not argue. Fewster and Verch use a threshold on the supremum of the scalar curvature to build order-valued functors on spacetimes, for a different purpose (Fewster and Verch, 2012, §4.3); the present rule is pointwise and is monotone along causal curves only on the subcategory.

Remark 21 (Local clocks)**.** More generally, let $\mathcal{C} \subseteq \mathsf{GH}$ be a full subcategory closed under restriction and containing $\mathbb{M}$, and let κ be a two-valued isometry invariant of germs of the metric that is non-decreasing along timelike curves in every member of $\mathcal{C}$ and non-constant on some member M. Then $I^- \cup \{(x,b) : \kappa(x) < \kappa(b)\}$ is a becoming theory on $\mathcal{C}$ that equals I^- on $\mathbb{M}$ and relates a spacelike pair of M, by Lemma 48. So Theorem 8(a) can hold on $\mathcal{C}$ only if $\mathcal{C}$ has no such

local clock. If disconnected spacetimes are admitted as objects, as in (Fewster and Verch, 2012, §§2.1–2.2), the converse holds too, and the theorem on $\mathcal{C}$ is equivalent to the statement that every germ of $\mathcal{C}$ occurs chronologically after and before a flat region, through finite chains of members of $\mathcal{C}$. That criterion is weaker than the pocket problem: no slab is transported. We do not know whether $\mathsf{GH}_{\mathrm{DEC}}$ satisfies it.

3.7 What the classification shows

The robust part of Stein's theorem. In general relativity, under the premises of Remark 3, nothing outside an event's causal past can have become for it unless everything has. This holds in every globally hyperbolic spacetime, however asymmetric, and does not rely on any foliation. Those who want an extended, spacelike "now" as part of objective becoming face Stein's dilemma in general relativity too, unless they change the category or reject locality on the metric-only category.

The strongest rival: clock-equipped spacetimes. Let GH_T have as objects pairs (M, T) with $M \in \mathsf{GH}$ and T a smooth Cauchy temporal function, and as morphisms $\psi : (M, T) \to (M', T')$ the morphisms of GH with $T' \circ \psi = T$. The rule $x \, B_{(M,T)} \, b \iff T(x) \leq T(b)$ is transitive, contains I^-, satisfies (LC) on this category, and has the level sets of T as shared presents. The Milne inclusion of Proposition 6 is not a morphism here: its clock $\tau_W = (t^2 - |\mathbf{x}|^2)^{1/2}$ has no smooth extension across the light cone, where its gradient diverges. So this rival is local with respect to its own morphisms. It rejects one premise, that becoming is a function of the metric alone over all of GH, and Theorem 8 then says nothing about it (in effect Monton's move). Its cost depends on the clock. A clock definable from the metric, as cosmological time is in big-bang cosmologies, adds no structure; the price is that the rule gives no verdict where that clock does not exist, and locality no longer selects it, since J^- also satisfies the axioms on GH_T. A preferred foliation or reference field not definable from the metric is added structure. Rejecting locality on GH itself is a distinct option (Proposition 6).

A trilemma. Suppose a shared present is understood as a set of distinct events each of which has become for each other, the sense in which Stein's dilemma is posed. Then Corollary 9 can be read as a trilemma for becoming theories, which are transitive by definition, defined from the metric alone: locality, a shared (extended) present, and non-trivial becoming cannot all be had. Defenders of a shared now reject locality on GH, reject transitivity or change the category; local becoming theorists get point-like presents, which recalls Stein's remark that the converse of Putnam's premise leads to "a peculiarly extreme (but pluralistic!) form of solipsism" (Stein, 1968, p. 18); and giving up non-triviality is the block. Other notions of a present, for instance a set of events none of which has become for another, are not addressed by the corollary. The corollary says nothing about which horn to take. Rejecting transitivity buys a definite thing in flat spacetime. By Corollary 12, a local symmetric relation among causally unrelated events of $\mathbb{M}$ is empty or is spacelike separation itself, so the only local shared present there is the whole elsewhere. Changing the category need not mean adding structure: a restriction of the admissible spacetimes can suffice (Proposition 20).

The fragile part. Uniqueness does not survive. Minkowskian symmetry forced a choice between I^- and J^-; general relativity leaves open how lightlike-separated events stand, and a rule may let curvature along the light ray decide. Clifton and Hogarth's debate over causal versus chronological

becoming (Clifton and Hogarth, 1995, pp. 364–365) thus acquires a third family of options; the choice between the endpoints is equivalent to a regularity requirement (Proposition 16).

Locality does the work symmetry did. In a single curved spacetime, definability-style invariance constrains almost nothing (Proposition 5); cosmological time defined by proper time from the past boundary violates only locality (Proposition 6), while cosmic time read off a curvature scalar is local and violates only the demand that the rule be a becoming theory on every globally hyperbolic geometry (Proposition 20). So the curved-spacetime case for "only the past has become" rests on the off-shell premise of Remark 4 together with a locality principle of the kind locally covariant field theory takes for granted and akin to Viglione's Causal Connection Condition (Viglione, 2026) (which is stricter: (LC) allows dependence on spacelike events inside the convex region). Rejecting locality is a coherent option, but it is the option of global, cosmological becoming, with its own costs (Dieks, 2006; Viglione, 2026).

Limits. The theorem concerns relations of becoming, not ontology; it does not decide between presentism, the growing block and eternalism. An eternalist can accept J^- as what has happened by an event. Reading B_M as existence from b is a further bridge, which nothing here supplies. The theorem assumes that becoming is defined for every globally hyperbolic geometry (Remark 4); Section 3.6 shows that dropping this premise is not routine. Spacetimes that are not globally hyperbolic are not treated beyond Proposition 18. The time orientation is an input, through (CB); the theorem shows that, given an orientation, only the past lobe can have become, but it does not explain the orientation. Nor does it address quantum nonlocality. Clifton and Hogarth's maxim that whatever can causally influence an event has become for it (Clifton and Hogarth, 1995, p. 364) needs care. Read as a claim about signalling, it puts J^- into B_M. Bell-type dependence on spacelike events is not signalling, so it is no counterexample to that reading. What it challenges is the stronger thought that any dynamical dependence on an event requires that event to have become. If the fundamental dynamics lets spacelike events contribute to bringing an event about, the stronger reading puts them into B_M, and Theorem 8 makes becoming universal. A non-trivial local becoming theory that holds the stronger reading therefore presupposes that such contribution is restricted to the causal past. That is a bet about the dynamics, not a theorem of relativity.

3.8 Prior work on becoming

Stein's theorem and its Clifton–Hogarth form are the direct predecessors (Stein, 1968, 1991; Clifton and Hogarth, 1995). Rakić (1997) takes an event-relative, transitive "realized with respect to" relation, argues that under her constraints it is not definable from the causal geometry of Minkowski spacetime, and adds it as a conservative primitive: in our terms, the added-structure option. Point-based presents (Müller, 2006) and the causal-diamond present of Arthur and Savitt (Arthur, 2006, 2019; Savitt, 2009, 2015), with Dorato's critique (Dorato, 2011) and Aames's application to Gödel universes (Aames, 2022), are conceptual proposals without a curved-spacetime classification. Earman (2008) builds worldline becoming in general relativity from causal pasts and notes that once the symmetries of Minkowski spacetime are lost, the Clifton–Hogarth definability results no longer apply (Earman, 2008, p. 151); he does not pursue a definability result. In flat spacetime, definability of sets from relativistic structure is governed by automorphism groups (Madarász et al., 2025).

The category and the naturality schema are those of Brunetti et al. (2003); the locality idea is theirs, and what is offered here is a classification of relation-valued assignments. The closest

structural precedent is the SPASs property of Fewster and Verch (2012): theories conforming to a common notion of sameness, one a subtheory of the other, that are isomorphic in one spacetime should be isomorphic in all. Theorem 8 has the same shape, one Minkowski verdict fixing the alternative everywhere, but its objects are relations between events and its mechanism is transitivity and (CB). Fewster and Verch show that the full class of locally covariant theories fails the property, and Theorem 15 is a counterpart: I^- and $I^- \cup N$ agree on $\mathbb{M}$ and differ elsewhere. Stein himself posed a cross-model question of a different kind, whether a cosmological model can be deformed to one with Gödel-type relations (Stein, 1970, p. 594). Clifton and Hogarth's remark quoted after Corollary 13 is an observation of difficulty, not a posed problem. The flat-spacetime counterpart of Corollary 9, that relativity leaves no shared present under its symmetries, is established (Callender, 2000; Saunders, 2002).

A citation sweep of Clifton and Hogarth (1995) and Stein (1991) (Semantic Scholar, 24 October 2026; repeated with OpenAlex, 25 October 2026), and reading of the works named here, found no prior extension of Stein's theorem to curved spacetime. This bounds, but does not certify, priority.

A modal route to a relativistic present is related. Müller's branching-based proposal (Müller, 2019) defines co-presentness from branching structure, and its Theorem 1 asserts non-trivial co-presentness on every history outside two featureless cases. Under the necessary nonempty-chain reading, a comb with a dense set of branch points on a real trunk satisfies that paper's Definition 6, is not featureless, and has identity co-presentness on its trunk. In the comb family with branch set D a proper subset of $\mathbb{R}$, all histories have non-trivial co-presentness if and only if D is not dense. That concerns an abstract order, not a relativistic manifold, and the later formulation of the theory lies outside the construction.

Part II. Determination on a fixed background

4 What the causal structure guarantees

Grant Part I. Does the unreality of b's future from b, on views that read having become as being real, make the future open from b? Openness in the relevant sense is a matter of what the past fixes, and that depends on the laws and on global causal structure. Geroch's domain of prediction supplies the tool (Geroch, 1977), and Manchak's work on prediction its limits (Manchak, 2008).

Definition 22. An event x is *guaranteed from* b if every past-inextendible causal curve from x meets $I^-_{\mathrm{s}}(b)$. Write $\mathrm{Det}(b)$ for the set of such events. Clearly $I^-_{\mathrm{s}}(b) \subseteq \mathrm{Det}(b)$.

$\mathrm{Det}(b)$ is a property of the spacetime and the event. To connect it with what the laws fix, let $\mathcal{L}$ be the set of solutions of some field equations on M. Say that x is *fixed from* b *under* $\mathcal{L}$, and write $x \in F_{\mathcal{L}}(b)$, if any two solutions in $\mathcal{L}$ that agree on $I^-_{\mathrm{s}}(b)$ agree at x. The equations, background and globally admissible solution class are held fixed in this comparison; they are inputs to the definition, not data an observer is assumed to have measured in the past. Here agreement means equality of the specified classical fields on the whole open region, not equality of a finite collection of detector records. Say that $\mathcal{L}$ has the *domain-of-dependence property* if, for every open $U \subseteq M$, any two solutions that agree on U agree at every event all of whose past-inextendible causal curves meet U. We assume this uniqueness property explicitly; it is not inferred merely from an operational no-signalling condition.

Remark 23 (What is guaranteed and what is fixed)**.** If $\mathcal{L}$ has the domain-of-dependence property and consists of continuous fields, then

$$\mathrm{Det}(b) \cup J^-(b) \subseteq F_{\mathcal{L}}(b),$$

since $J^-(b)$ is the closure of $I^-_{\mathrm{s}}(b)$ in a globally hyperbolic spacetime (Minguzzi and Sánchez, 2008). So $\mathrm{Det}(b)$ is fixed under every such law: it is what the causal structure guarantees, whatever the fields. The inclusion is in general strict, and nothing about $F_{\mathcal{L}}(b)$ beyond it follows from $\mathrm{Det}(b)$ being small. For example, on Minkowski spacetime with the unit timelike field $u = \partial_t$, the local equations $u \cdot \nabla Q = P$, $u \cdot \nabla P = 0$ have the solutions $Q = A(\vec{X}) + tB(\vec{X})$, $P = B(\vec{X})$ for arbitrary smooth A, B. The values of (Q, P) at one point of an integral curve fix them along it, and the past integral curve from x is a past-inextendible causal curve, so the law has the domain-of-dependence property; and every integral curve of u enters $I^-_{\mathrm{s}}(b)$, so $F_{\mathcal{L}}(b) = M$ for every b. A law of this kind uses background structure beyond the metric, but nothing in the domain-of-dependence property excludes it. Neither notion is Belnap's settledness, on which a proposition is settled at a point event if it holds in every possible history through it (Belnap, 1992); under indeterministic laws they come apart from it, and we say nothing about such laws. We use $I^-_{\mathrm{s}}(b)$, which every local becoming theory counts as real from b, rather than $J^-(b)$; the characterization below uses that $I^-_{\mathrm{s}}(b)$ is open.

Relation to Geroch. Geroch's domain of prediction $P(q)$ is the set of x such that every past-directed timelike curve from x without past endpoint enters $I^-(q)$, but $I^-(x) \not\subseteq I^-(q)$ (Geroch, 1977, sec. 3). $\mathrm{Det}(b)$ is the causal-curve analogue of the union of $P(b)$ with the events whose pasts lie in that of b; we expect the two to differ at most on null boundaries, but have not proved it. Geroch gives an equivalent form by achronal surfaces in $I^-(q)$ (Geroch, 1977, sec. 3); Theorem 24 is a slicing form for globally hyperbolic spacetimes.

Theorem 24. *Let $t : M \to (\alpha, \beta)$ be a Cauchy time function. Then $x \in \mathrm{Det}(b)$ if and only if there is $c \in (\alpha, t(x))$ with $J^-(x) \cap S_c \subseteq I^-_{\mathrm{s}}(b)$.*

Proof. ($\Leftarrow$) A past-inextendible causal curve from x crosses the Cauchy surface S_c, at a point of $J^-(x) \cap S_c \subseteq I^-_{\mathrm{s}}(b)$.

($\Rightarrow$) Suppose that for every $c \in (\alpha, t(x))$ the set $K_c = (J^-(x) \cap S_c) \setminus I^-_{\mathrm{s}}(b)$ is nonempty; it is compact, since $I^-_{\mathrm{s}}(b)$ is open. Fix $t(x) > c_1 > c_2 > \dots$ with $c_n \to \alpha$, and let

$$P_n = \{(z_1, \dots, z_n) \in K_{c_1} \times \cdots \times K_{c_n} : x \geq z_1 \geq z_2 \geq \cdots \geq z_n\}.$$

P_n is compact, since $\leq$ is closed. It is nonempty: pick $y \in K_{c_n}$ and a causal curve from x to y; it crosses each S_{c_i}, $i < n$, at some $z_i \leq x$, and $z_i \notin I^-_{\mathrm{s}}(b)$, for otherwise $y \leq z_i \ll b$ would give $y \ll b$. The maps $P_{n+1} \to P_n$ forgetting the last coordinate are continuous, so the inverse limit of the nonempty compact spaces P_n is nonempty, giving $x \geq z_1 \geq z_2 \geq \cdots$ with $z_n \in K_{c_n}$. Join consecutive terms by causal curves and concatenate. Each segment avoids $I^-_{\mathrm{s}}(b)$ by push-up, as before. Along the concatenation $t \to \alpha$, so it has no past endpoint in M: it is a past-inextendible causal curve from x missing $I^-_{\mathrm{s}}(b)$. □

Corollary 25. *If $I^-_{\mathrm{s}}(b)$ contains a Cauchy surface S, then $\mathrm{Det}(b) = M$.*

Proof. If $x \in I^-_{\mathrm{s}}(S) \cup S$, then $x \leq s \ll b$ for some $s \in S$, so $x \in I^-_{\mathrm{s}}(b)$. If $x \in I^+_{\mathrm{s}}(S)$, extend a past-inextendible causal curve from x to an inextendible one; it meets S, and not to the future of x, since then $s' \ll x \leq s$ for some $s, s' \in S$, contradicting achronality. So the curve from x meets $S \subseteq I^-_{\mathrm{s}}(b)$. □

Since $\mathrm{Det}(b)$ is defined by causal curves alone, it depends only on the conformal class of the metric; conformal diagrams decide it.

5 The standard models

Lemma 26. *Let (Σ, d) be $\mathbb{R}^3$, hyperbolic space H^3, or the unit sphere S^3 with $r_2 \le \pi$. Then the closed ball $\bar{B}(X, r_1)$ lies in the open ball $B(Y, r_2)$ if and only if $d(X, Y) + r_1 < r_2$.*

Proof. ($\Leftarrow$) is the triangle inequality. ($\Rightarrow$) Let $r_1 \ge 0$ and suppose $d(X, Y) + r_1 \ge r_2$. On $\mathbb{R}^3$ and H^3, where geodesics extend indefinitely and minimize, or on S^3 when $d(X, Y) + r_1 \le \pi$, follow a minimizing geodesic from Y through X (any geodesic from X if $X = Y$) a further distance r_1; the endpoint lies in $\bar{B}(X, r_1)$ at distance $d(X, Y) + r_1 \ge r_2$ from Y. If $d(X, Y) + r_1 > \pi$ on S^3, the antipode of Y is at distance $\pi - d(X, Y) < r_1$ from X and $\pi \ge r_2$ from Y. □

Proposition 27. *Let M be conformal to $(\eta_-, \eta_+) \times \Sigma$ with metric $-d\eta^2 + d\Sigma^2$, where Σ is $\mathbb{R}^3$, H^3, or the unit S^3 with $\eta_+ - \eta_- \le \pi$. Then $\mathrm{Det}(b) = I_{\mathrm{s}}^-(b)$ for every b.*

Proof. Use $t = \eta$. For $x = (\eta_x, X)$, $b = (\eta_b, Y)$: $x \ll b$ iff $d(X, Y) < \eta_b - \eta_x$ (on S^3, $\eta_b - \eta_x < \pi$). For $\eta_- < c < \eta_x$, $J^-(x) \cap S_c = \bar{B}(X, \eta_x - c)$ and $I_{\mathrm{s}}^-(b) \cap S_c = B(Y, \eta_b - c)$, with $\eta_b - c < \pi$ on S^3. By Lemma 26 the inclusion holds iff $d(X, Y) < \eta_b - \eta_x$, i.e. iff $x \ll b$. Apply Theorem 24. □

This covers Minkowski spacetime, spatially flat and open FLRW universes with a big bang (conformal to $\eta > 0$), the closed radiation universe ($a \propto \sin\eta$, $\eta \in (0, \pi)$) and global de Sitter spacetime ($\Sigma = S^3$, $\eta \in (-\pi/2, \pi/2)$) (Hawking and Ellis, 1973). In none of them is anything outside an event's chronological past guaranteed from it. In Minkowski spacetime and the spatially flat universes this bound is attained by a natural field.

Proposition 28. *Let $M = (\eta_-, \eta_+) \times \mathbb{R}^3$ with $g = a(\eta)^2(-d\eta^2 + d\vec{X}^2)$, $a > 0$ smooth, and let $\mathcal{L}$ be the smooth solutions of the conformally coupled scalar equation $\Box_g \phi - \frac{1}{6} R\phi = 0$. Then $F_{\mathcal{L}}(b) = J^-(b)$ for every b.*

Proof. $J^-(b) \subseteq F_{\mathcal{L}}(b)$ by continuity, as in Remark 23. Conversely, ϕ is a solution iff $\psi = a\phi$ solves the flat wave equation $\partial_\eta^2 \psi = \Delta\psi$ on M (Wald, 1984, app. D). Let $b = (\eta_b, \vec{Y})$ and $x = (\eta_x, \vec{X}) \notin J^-(b)$, so $\eta_x - \eta_b + |\vec{X} - \vec{Y}| > 0$, since the causal curves of M are those of the conformal Minkowski metric and the slab is causally convex in Minkowski spacetime. Choose a unit vector $\hat{k}$ with $\hat{k} \cdot (\vec{X} - \vec{Y}) = -|\vec{X} - \vec{Y}|$, a smooth f vanishing on $(-\infty, 0]$ and positive on $(0, \infty)$, and put $\psi = f(u)$ with $u = \eta - \eta_b - \hat{k} \cdot (\vec{Z} - \vec{Y})$ at $(\eta, \vec{Z})$. This solves the wave equation. On $I_{\mathrm{s}}^-(b)$, $\eta - \eta_b < -|\vec{Z} - \vec{Y}|$, so $u < 0$ and $\psi = 0$; at x, $u > 0$ and $\psi \ne 0$. So ψ/a and 0 are solutions that agree on $I_{\mathrm{s}}^-(b)$ but not at x. □

Remark 29 (The class matters)**.** The witness in this proof is a plane wave, which has infinite energy. The verdict $F_{\mathcal{L}}(b) = J^-(b)$ belongs to the class of all smooth solutions as well as to the equation. The next proposition shows the verdict changing with the class for one equation on one spacetime.

Proposition 30 (Massive fields on Minkowski spacetime)**.** *On Minkowski spacetime $\mathbb{R}^{1+n}$ let $m > 0$ and consider real solutions of $\partial_t^2 \phi - \Delta\phi + m^2\phi = 0$.*

(a) For the class of all smooth solutions, $F(b) = J^-(b)$ for every b.

(b) *For $n = 1$ and the class of finite-energy solutions (Cauchy data in $H^1 \times L^2$), $F(b)$ is the whole spacetime for every b, and the same holds with the strict chronological future of b as base.*

Proof. (a) $J^-(b) \subseteq F(b)$ by continuity. Let $x \notin J^-(b)$, with $b = (t_b, \vec{Y})$. As in the proof of Proposition 28 choose a unit vector $\hat{k}$ such that $u = t - t_b - \hat{k} \cdot (\vec{Z} - \vec{Y})$ is negative on $I_{\mathrm{s}}^-(b)$ and positive at x, and put $v = t - t_b + \hat{k} \cdot (\vec{Z} - \vec{Y})$. For functions of (u, v) the equation reads $4\phi_{uv} + m^2\phi = 0$. The function $j(z) = J_0(m\sqrt{z})$ is entire and satisfies $zj'' + j' + \frac{m^2}{4} j = 0$, so $R = j((u-s)(v-c))$ satisfies $R_{uv} = -\frac{m^2}{4} R$ for all real s, c. Let $h \geq 0$ be smooth with support in $(0, \infty)$ and put

$$\phi(u, v) = \int_0^u h(s)\, j\big((u-s)(v-c)\big)\, ds \quad (u > 0), \qquad \phi = 0 \quad (u \leq 0).$$

Since $R = 1$ at $s = u$, $\phi_u = h(u) + \int_0^u h R_u \, ds$, and so $\phi_{uv} = \int_0^u h R_{uv} \, ds = -\frac{m^2}{4}\phi$. Thus ϕ is a smooth solution vanishing on $I_{\mathrm{s}}^-(b)$. Taking c close to $v(x)$ and h a bump supported below $u(x)$ gives $\phi(x) > 0$.

(b) Take b at the origin and let ϕ have finite energy and vanish on $I_{\mathrm{s}}^-(b) = \{t < -|x|\}$. At each $t < 0$ the energy density vanishes on $|x| < |t|$, so all of the conserved energy lies in $|x| \geq |t|$. For a finite-energy solution that portion tends to zero as $t \to -\infty$ (Burnol, 2006, Theorem 1, with time reversed). Hence the energy is zero, and $\phi = 0$ because $m > 0$. The future base is the time reverse. □

The solutions used in (a) grow like a modified Bessel function where $(u-s)(v-c) < 0$ and have infinite energy. The argument for (b) uses only the decay of energy outside the light cone, which Burnol notes follows from Hörmander's estimates and fails at zero mass; for $n \geq 2$ we have not checked a source for it. A massive field carries nothing in from null infinity: its energy arrives from past timelike infinity, which lies in the closure of every past light cone. So on this background the past of an event fails to fix a finite-energy field only when the field is massless. Weinstein argues that prediction from a past light cone in electromagnetism needs the added assumption of no incoming radiation (Weinstein, 2011); part (b) says that for a massive field of finite energy the corresponding statement is a theorem, and a time-symmetric one. The fixing in (b) uses the arbitrarily remote past, so it does not make such a field practically predictable (compare Section 6).

Proposition 31. *Let M be conformal to $(\eta_-, \eta_+) \times S^3$ with $\eta_+ - \eta_- > \pi$. Then* $\mathrm{Det}(b) = I_{\mathrm{s}}^-(b)$ *if $\eta_b \leq \eta_- + \pi$, and* $\mathrm{Det}(b) = M$ *if $\eta_b > \eta_- + \pi$. In the Einstein static universe ($\eta_\pm = \pm\infty$),* $\mathrm{Det}(b) = M$ *for every b.*

Proof. If $\eta_b > \eta_- + \pi$, take $c \in (\eta_-, \eta_b - \pi)$: $I_{\mathrm{s}}^-(b) \cap S_c$ is an open ball of radius greater than π, hence all of S_c, and Corollary 25 applies. If $\eta_b \leq \eta_- + \pi$, every $c > \eta_-$ has $\eta_b - c < \pi$, and the proof of Proposition 27 applies. □

For the closed dust FLRW model, $a(\eta) = \frac{1}{2} a_m (1 - \cos\eta)$ with $\eta \in (0, 2\pi)$, the threshold is $\eta = \pi$, the moment of maximum expansion. That the particle horizon of the closed dust model disappears at maximum expansion, so that an observer can then in principle see the whole universe, is known (Grøn and Johannesen, 2011; Bolotin and Tanatarov, 2013); the Einstein static case is Geroch's (Geroch, 1977, Fig. 5). What Proposition 31 adds is the consequence for determination: with Corollary 25, once the particle horizon has vanished ($\eta_b > \pi$), everything is guaranteed from b,

so that every law with the domain-of-dependence property fixes everything from what is real from b. This fits Manchak's theorem that if some event of b's timelike future lies in Geroch's domain of prediction of b, the spacetime admits a compact spacelike slice (Manchak, 2008, p. 319). That theorem concerns the geometric notion. The flow law of Remark 23 shows that it has no analogue for what a particular law fixes: in Minkowski spacetime, which has no compact slice, that law fixes all of b's timelike future. Manchak also records the view that "Maxwell's equations constrain some spacetimes in such a way as to allow for prediction" (Manchak, 2008, p. 321 and n. 15), so the law-relative notion has antecedents. Earman's discussion explicitly includes the Cauchy-surface exception and an electromagnetic-constraint example citing Ellis and Sciama (Earman, 1995, pp. 125–130). We have checked these selected original Earman pages; Ellis and Sciama (1972, pp. 41–42) remain unread here and enter this comparison through Earman's discussion. The model-specific characterization below may overlap with those sources; we make no priority claim for it.

Before maximum expansion a particular field can fix more than is guaranteed. Light leaving a point of the closed model refocuses at its antipode after conformal time π, and the rescaled conformal scalar $\psi = a\phi$ returns there with its sign reversed.

Proposition 32. *Let M be the closed dust model, $g = a(\eta)^2(-d\eta^2 + d\Omega_3^2)$ with $a = \frac{1}{2}a_m(1-\cos\eta)$, $\eta \in (0, 2\pi)$, and $\mathcal{L}$ as in Proposition 28. Let $A(\eta, X) = (\eta + \pi, -X)$, where $-X$ is the antipode of X. Then $J^-(b) \cup (A(J^-(b)) \cap M) \subseteq F_{\mathcal{L}}(b)$. The set $A(J^-(b)) \cap M$ is disjoint from $I_{\mathrm{s}}^+(b)$, and if $\eta_b \leq \pi$ it is nonempty and disjoint from* $\mathrm{Det}(b)$.

Proof. Here $\psi = a\phi$ solves the conformally invariant equation of the Einstein static universe, $\partial_\eta^2\psi = \Delta\psi - \psi$ on the unit S^3 (Wald, 1984, app. D), and by global well-posedness extends uniquely to $\mathbb{R} \times S^3$. Expanding in spherical harmonics of degree n, which have frequency $n+1$ and parity $(-1)^n$ under the antipodal map, each mode, and so every smooth solution, satisfies $\psi(\eta + \pi, -X) = -\psi(\eta, X)$. Two solutions that agree on $I_{\mathrm{s}}^-(b)$ agree, by continuity, on $J^-(b)$, and hence on $A(J^-(b)) \cap M$. If $(\eta, X) \in J^-(b)$ and $\eta + \pi < 2\pi$, then $d(X, Y) \leq \eta_b - \eta$, so $d(-X, Y) = \pi - d(X, Y) \geq (\eta + \pi) - \eta_b$; thus $A(\eta, X) \notin I_{\mathrm{s}}^+(b)$. If $\eta_b \leq \pi$, then $\mathrm{Det}(b) = I_{\mathrm{s}}^-(b)$ by Proposition 31, which lies in $\{\eta < \eta_b\}$, while $A(J^-(b))$ lies in $\{\eta > \pi\}$ and meets M, for instance at $A(\eta_b/2, Y)$ when $b = (\eta_b, Y)$. □

For $\eta_b \leq \pi$, the echo events are unreal from b and fixed from b, but they lie beside b or on its future light cone, not in its timelike future. The field fixes more than the echo, including part of b's timelike future. Let $\Diamond_b = \{(\eta, X) \in \mathbb{R} \times S^3 : d(X, Y) + |\eta| < \eta_b\}$, the double cone in the Einstein static universe between b and its mirror image $(-\eta_b, Y)$, and $U_b = \Diamond_b \cup A\Diamond_b \cup A^2\Diamond_b$.

Theorem 33. *In the setting of Proposition 32, let $0 < \eta_b < \pi$. Then $F_{\mathcal{L}}(b) = M \cap \overline{U_b}$. In particular $F_{\mathcal{L}}(b)$ contains the nonempty set $\{(\eta, X) \in M : d(X, Y) \leq \eta + \eta_b - 2\pi\}$, which lies in $I_{\mathrm{s}}^+(b)$.*

Proof. For data (f, g) at $\eta = 0$, the Einstein static solution is $\psi(\eta, X) = \partial_\eta[\sin\eta\, M_f(X, \eta)] + \sin\eta\, M_g(X, \eta)$, where $M_h(X, r)$ is the mean of h over the geodesic sphere of radius r about X; one checks this on each mode, using that the zonal function of degree n on S^3 is $\sin((n+1)r)/((n+1)\sin r)$. So $\psi(x)$ depends on the data only near the sphere $\Sigma_x = S(X, \eta_x)$, which is the point $-X$ when $\eta_x = \pi$.

Sufficiency. Solutions agreeing on $I_{\mathrm{s}}^-(b)$ agree on the ball $B(Y, \eta_b - \epsilon)$ in the slice $\eta = \epsilon$, hence on its domain of dependence in the Einstein static universe; the union over $\epsilon > 0$ is $\Diamond_b$. The identities $\psi \circ A = -\psi$ and $\psi \circ A^2 = \psi$ carry the agreement to $A\Diamond_b$ and $A^2\Diamond_b$. Moreover, $F_{\mathcal{L}}(b)$

is relatively closed in M: it is the intersection of the zero sets of all smooth difference solutions vanishing on $I_{\rm s}^-(b)$. Since M is open in the Einstein cylinder, taking the relative closure gives $M \cap \overline{U_b} \subseteq F_{\mathcal{L}}(b)$.

Necessity. Let $x \in M \setminus \overline{U_b}$. Replacing x by $A^{-1}x$ if $\eta_x > \pi$, assume $\eta_x \leq \pi$. Let $C = \bar{B}(Y, \eta_b)$. Data supported off C give a solution vanishing on $I_{\rm s}^-(b)$: if $z \in I_{\rm s}^-(b)$ lay in its support, then $w \leq z \ll b$ for some w in the support of the data, so $d(w, Y) < \eta_b$. If $\eta_x = \pi$, then $d(-X, Y) > \eta_b$, and data $(f, 0)$ with f a bump at $-X$ off C give $\psi(x) = -f(-X) \neq 0$. If $\eta_x < \pi$, then $\Sigma_x \not\subseteq C$: otherwise the connected set $S^3 \setminus C$ lies in one of the two open balls bounded by Σ_x, so the other closed ball lies in C, which by the argument of Lemma 26 means $d(X, Y) + \eta_x \leq \eta_b$ or $d(-X, Y) + \pi - \eta_x \leq \eta_b$, that is, $x \in \overline{\Diamond_b} \cup \overline{A\Diamond_b}$. Take $p \in \Sigma_x \setminus C$ and data $(0, g)$ with $g \geq 0$ supported near p off C and $g(p) > 0$; then $\psi(x) = \sin \eta_x \, M_g(X, \eta_x) > 0$.

Finally, if $d(X, Y) \leq \eta + \eta_b - 2\pi$, then $d(X, Y) < \eta - \eta_b$ because $\eta_b < \pi$, so $(\eta, X) \in I_{\rm s}^+(b)$; the set contains (η, Y) for $\eta \geq 2\pi - \eta_b$. □

On b's comoving geodesic $X = Y$, the field is fixed from b exactly for $\eta \leq \eta_b$ and for $\eta \geq 2\pi - \eta_b$, and open in between: the end is fixed before the middle. The reason is that the past light cone of an event just before the crunch wraps around the universe and refocuses near Y at the big bang, inside what is real from b. The same sphere argument shows that, up to boundary, $M \cap U_b$ is the set of events all of whose past-inextendible null geodesics meet $I_{\rm s}^-(b)$, so for this field, which propagates on light rays alone, the null geodesics decide what is fixed. The same argument applies to global de Sitter spacetime, conformal to the slab $(-\pi/2, \pi/2)$, where the field is fixed from b on the antipodal comoving geodesic for all $\eta > -\eta_b$, although no signal from that geodesic ever reaches the comoving geodesic of b. So the geometric threshold at maximum expansion marks when everything is guaranteed, not when this field first fixes anything beyond the past.

For a law whose fixed set itself changes at the threshold, take the comoving flow $u = a^{-1}\partial_\eta$ of the closed dust model with the equations of Remark 23; here u is the dust's own four-velocity, so the law adds no structure the model lacks. The comoving line through X meets $I_{\rm s}^-(b)$ iff $d(X, Y) < \eta_b$, so, by continuity in X and by choosing data supported off the closed ball, $F_{\mathcal{L}}(b) = (0, 2\pi) \times \bar{B}(Y, \eta_b)$. This is a proper subset of M, containing events in b's timelike future, if $\eta_b < \pi$, and all of M if $\eta_b \geq \pi$. Under this law the answer to "is the future fixed from here?" differs between events before and after maximum expansion.

Theorem 35 shows that on closed dust a local law can fix exactly $\mathrm{Det}(b) \cup J^-(b)$ for every $\tau \neq \pi$. Whether some local law does so at $\tau = \pi$, or in other closed models, we leave open.

6 Changing the law

Write $\tau = \eta_b$ for the conformal time of a standpoint.

The massless example is special. Its extra fixing can change even when the metric and its causal curves are held fixed. Let $\mathcal{L}_m$ consist of all smooth real solutions on the closed dust strip of

$$(\Box_g - R_g/6 - m^2)\phi = 0,$$

with ordinary constant physical mass $m \geq 0$, and write $F_m(b)$ for its fixing set. Set $\Lambda^2 = -\Delta_{S^3} + 1$ and $T = 2\pi$. With $\psi = a\phi$ and $\lambda = m^2$, the rescaled equation is

$$\psi_{\lambda,\eta\eta} + (\Lambda^2 + \lambda q(\eta))\psi_\lambda = 0, \qquad q(\eta) = a(\eta)^2. \tag{1}$$

In particular a constant addition to the rescaled potential would be a different perturbation, corresponding to physical potential δ/a^2.

Proposition 34 (Loss of exact fixing under ordinary mass)**.** *Fix the dust amplitude $a_m > 0$. For each $b = (\tau, Y)$ with $0 < \tau < \pi$, there are a fixed event $x_* = (T - s, Y)$, $0 < s < \tau$, and $m_* > 0$ such that*

$$x_* \in F_0(b) \cap I_{\mathrm{s}}^+(b), \qquad x_* \notin F_m(b) \quad \textit{for every } 0 < m < m_*.$$

The background metric is the same for all these solution classes.

Proof. Choose a nonzero smooth $v \geq 0$ supported strictly outside $\bar{B}(Y, \tau)$, and solve (1) with data $(0, v)$ at the auxiliary cylinder slice $\eta = 0$. The coefficient $q = a^2$ is smooth through 0 and T, so ordinary smooth Cauchy evolution and parameter dependence apply on the extended cylinder. This is a device for constructing solutions on the open physical strip, not a physical continuation through its singularities. For $z = (\eta, Z) \in I_{\mathrm{s}}^-(b)$, the initial ball $\bar{B}(Z, \eta)$ lies in $B(Y, \tau)$ by the triangle inequality. Finite propagation therefore gives $\psi_\lambda = 0$ throughout $I_{\mathrm{s}}^-(b)$ for every λ; the zero-order term does not change this conclusion.

Put $S_0(u) = \sin(u\Lambda)/\Lambda$. The first variation $w = \partial_\lambda \psi_\lambda|_{\lambda=0}$ has zero initial data and source $-q(u)S_0(u)v$. Since $S_0(T - u) = -S_0(u)$, Duhamel's formula gives

$$w(T) = \int_0^T q(u) S_0(u)^2 v \, du.$$

The frequencies are $k = n + 1$. Expanding $q = (a_m^2/4)(3/2 - 2\cos u + \frac{1}{2}\cos 2u)$ gives

$$\int_0^T q(u) \frac{\sin^2(ku)}{k^2} \, du = \begin{cases} 5\pi a_m^2/16, & k = 1, \\ 3\pi a_m^2/(8k^2), & k \geq 2. \end{cases}$$

Thus, with Π_0 the projection onto spatial constants,

$$w(T) = \left(\frac{3\pi a_m^2}{8} \Lambda^{-2} - \frac{\pi a_m^2}{16} \Pi_0 \right) v.$$

The kernel of Λ^{-2} on the unit S^3 is $G(r) = (\pi - r)/(4\pi^2 \sin r)$: it solves the homogeneous radial equation away from its pole, is smooth at the antipode, and has the normalized singularity $1/(4\pi r)$. Since Π_0 has kernel $1/(2\pi^2)$, the kernel of the operator displayed above is

$$K(r) = \frac{a_m^2}{32\pi} \left(\frac{3(\pi - r)}{\sin r} - 1 \right) > 0 \qquad (0 < r \leq \pi),$$

using $(\pi - r)/\sin r \geq 1$, with its continuous value at $r = \pi$. Consequently $w(T, Y) > 0$. Continuity now permits a choice of a fixed $s \in (0, \tau)$ with $w(T - s, Y) > 0$. Theorem 33 puts $x_* = (T - s, Y)$ in $F_0(b) \cap I_{\mathrm{s}}^+(b)$, so $\psi_0(x_*) = 0$. Smooth dependence on λ yields

$$\psi_{m^2}(x_*) = m^2 w(x_*) + o(m^2) > 0$$

for every sufficiently small positive m. Because $a(x_*) > 0$, the physical field and zero agree on $I_{\mathrm{s}}^-(b)$ but differ at x_*. The event and data were chosen before the mass interval. □

Theorem 35 below gives the full formula; this proposition supplies an explicit first-order witness. The result concerns a linear classical field on a fixed background, without interactions, quantization or gravitational backreaction.

Theorem 35 (Massive fields, complete classification)**.** *On the closed dust model, for each $m > 0$ let $\mathcal{L}_m$ be all smooth real solutions of $(\Box_g - R_g/6 - m^2)\phi = 0$, and let $b = (\tau, Y)$. Then*

$$F_m(b) = J^-(b) \quad \text{if } 0 < \tau < \pi, \qquad F_m(b) = M \quad \text{if } \pi \le \tau < 2\pi.$$

The second line also holds for $m = 0$.

Outline. Put $\psi = a\phi$, so that $\psi_{\eta\eta} + (\Lambda^2 + V(\eta))\psi = 0$ with $V = m^2a^2$ entire and positive on $(0, 2\pi)$. A successive-approximation construction gives a line Riemann function jointly analytic in its variables. Periodizing it gives the displacement and velocity kernels on S^3. The analytic pieces of the kernels are not identically zero, so the joint support of the two Cauchy kernels has no open holes. For $\tau < \pi$ and $x \notin J^-(b)$, the kernel support therefore leaves the closed ball $\bar{B}(Y, \tau)$, and data supported off that ball give a solution vanishing on $I_{\mathrm{s}}^-(b)$ and not at x; the reverse inclusion is continuity. For $\tau > \pi$ the trace of $I_{\mathrm{s}}^-(b)$ on an early slice is the whole sphere. For $\tau = \pi$ it is the sphere minus a point, and smoothness forces both data to vanish. The full argument is in the project's public note (Appendix C). □

So before maximum expansion any positive mass removes every island of extra fixing, while the threshold itself does not move, and at the exact threshold $\tau = \pi$ the law fixes everything although $\mathrm{Det}(b) = I_{\mathrm{s}}^-(b)$. Dense traces do this work there, not causal curves.

6.1 A norm bound for hidden alternatives

Exact fixing quantifies over every admissible difference, however small. To ask about its size, a restriction on alternatives must be supplied. Use the conformal Cauchy-data norm

$$N(f, v)^2 = \|\Lambda f\|_2^2 + \|v\|_2^2$$

at the auxiliary slice $\eta = 0$, with unit-S^3 volume $2\pi^2$. This is not an observer's measured physical-energy budget.

Proposition 36. *Let ψ_λ and ψ_0 have the same smooth data (f, v) and solve the cylinder equation with spatially homogeneous smooth real potential $\lambda q(\eta)$ and with zero potential, respectively. For $\eta \ge 0$, put $Q(\eta) = \int_0^\eta |q(u)|\, du$. Then*

$$\sup_X |\psi_\lambda(\eta, X) - \psi_0(\eta, X)| \le \frac{|\lambda| Q(\eta) e^{|\lambda| Q(\eta)}}{\sqrt{6}}\, N(f, v). \tag{2}$$

If $f = 0$, the denominator improves to $\sqrt{12}$.

Proof. Let $C_{\lambda,k}$ and $S_{\lambda,k}$ be the fundamental solutions with data $(1, 0)$ and $(0, 1)$ at frequency $k \ge 1$. Their Volterra equations and Gronwall's inequality imply $|C_{\lambda,k}| \le e^{|\lambda| Q/k}$ and $|S_{\lambda,k}| \le e^{|\lambda| Q/k}/k$. Substitution back into those equations bounds the differences from the free multipliers by

$$|C_{\lambda,k} - \cos k\eta| \le B/k, \qquad |S_{\lambda,k} - \sin(k\eta)/k| \le B/k^2, \quad B = |\lambda| Q e^{|\lambda| Q}.$$

Acting on Λf rather than f, the first multiplier acquires another $1/k$. The addition identity for an orthonormal harmonic basis is $\sum_j |Y_{kj}(X)|^2 = k^2/(2\pi^2)$, since the eigenspace has dimension k^2 and its diagonal projector is rotation-invariant. Joint Cauchy–Schwarz on the coefficients of $(\Lambda f, v)$ bounds the squared evaluation norm by

$$\sum_{k=1}^{\infty} \frac{k^2}{2\pi^2} \left(\frac{B^2}{k^4} + \frac{B^2}{k^4} \right) = \frac{B^2}{6}.$$

Removing the position-data term halves this bound. The argument concerns smooth solutions and the bounded difference functional; it does not assign point values to arbitrary individual finite-energy fields. □

Apply this to any rescaled difference $h_m = a(\phi_1 - \phi_2)$ of two massive solutions agreeing exactly on $I_s^-(b)$, with data (f, v). Smooth extension to zero implies that both data vanish on $B(Y, \tau)$. The massless solution h_0 with the same data vanishes on $I_s^-(b)$ by finite propagation. Thus $h_0(x) = 0$ for every $x \in F_0(b)$, and (2) gives

$$|h_m(\eta, X)| \le \frac{m^2 Q(\eta) e^{m^2 Q(\eta)}}{\sqrt{6}} N(f, v), \qquad (\eta, X) \in F_0(b), \tag{3}$$

where, for dust,

$$Q(\eta) = \frac{a_m^2}{4}\left(\frac{3\eta}{2} - 2\sin\eta + \frac{\sin 2\eta}{4}\right).$$

The bound controls all such smooth alternatives with $N \le 1$, not just the witness in Proposition 34. At fixed targets their rescaled ambiguity is $O(m^2)$. For physical fields divide by $a(\eta) > 0$; uniform physical-field control requires staying away from the singular endpoints. Without a norm bound any nonzero hidden solution can instead be scaled to an arbitrarily large target difference. Neither convention is an observation model supplied by the geometry.

6.2 Errors in records are a different question

Equation (3) retains exact agreement on the whole past while changing the law. It is not a stability theorem for noisy records under a fixed law. Even at a massless-fixed target, an aggregate field-error norm need not control a point value. To see the regularity issue explicitly, fix any interior target (η_*, Y) and let Z_k be the real unit-L^2 zonal harmonic centered at Y, with $Z_k(Y) = k/\sqrt{2\pi^2}$. The smooth massless solutions

$$\psi_K(\eta, X) = \frac{1}{\sqrt{K}} \sum_{k=K}^{2K-1} \frac{\cos(k(\eta - \eta_*))}{k} Z_k(X)$$

have conserved conformal norm $N = 1$ and target value $\sqrt{K/(2\pi^2)}$, but

$$\|\psi_K\|^2_{L^2((0,2\pi)\times S^3)} = \frac{\pi}{K} \sum_{k=K}^{2K-1} \frac{1}{k^2} \le \frac{\pi}{K^2}.$$

Their error on $I_s^-(b)$ is even smaller. They do not agree *exactly* there. This is unbounded point evaluation under a specified norm, already present with records of the entire strip, not a causal-horizon or detector failure. Every fixed spatial average with profile $\chi \in L^2(S^3)$ instead satisfies $|\langle \chi, \psi_K(\eta)\rangle| \le \|\chi\|_2 / K$. No general past-record stability theorem for such averages is asserted.

A common finite spectral cutoff changes the exact question too: every solution in that restricted massless class is a finite real-analytic sum, so agreement on any nonempty open region entails agreement everywhere. Its fixing set is all of M. Thus finite mode sums cannot establish exact past agreement with a nonzero future difference in the unrestricted smooth theory. These are classical regularity observations, not further priority claims or results about the physical measurement process.

7 The fixing map

The results combine into a single picture. For a law class $\mathcal{L}$ of continuous fields with the domain-of-dependence property, and a non-universal becoming theory,

$$I_{\mathrm{s}}^-(b) \;\subseteq\; B(b) \;\subseteq\; J^-(b) \;\subseteq\; F_{\mathcal{L}}(I_{\mathrm{s}}^-(b)), \qquad I_{\mathrm{s}}^-(b) \;\subseteq\; \mathrm{Det}(b) \;\subseteq\; F_{\mathcal{L}}(I_{\mathrm{s}}^-(b)). \tag{4}$$

Theorem 37 (All patterns occur)**.** *Each row below holds, and each of* $J^-(b) \subseteq F$, $I_{\mathrm{s}}^-(b) \subseteq \mathrm{Det}(b)$ *and* $\mathrm{Det}(b) \subseteq F$ *can be strict or an equality. Here* $B(b) = \{x : x \, B_M \, b\}$ *and* $F = F_{\mathcal{L}}(I_{\mathrm{s}}^-(b))$.

Spacetime, law, standpoint	$\mathrm{Det}(b)$	$F(b)$
Minkowski, conformal scalar	$I_{\mathrm{s}}^-(b)$	$J^-(b)$
Minkowski, preferred-flow law	$I_{\mathrm{s}}^-(b)$	M
Closed dust, $\tau < \pi$*, conformal scalar*	$I_{\mathrm{s}}^-(b)$	$M \cap \overline{U_b}$*, three diamonds*
Closed dust, $\tau < \pi$*, comoving flow law*	$I_{\mathrm{s}}^-(b)$	$(0, 2\pi) \times \bar{B}(Y, \tau)$
Closed dust, $\tau < \pi$*, mass* $m > 0$	$I_{\mathrm{s}}^-(b)$	$J^-(b)$
Closed dust, $\tau = \pi$*, mass* $m \geq 0$	$I_{\mathrm{s}}^-(b)$	M
Closed dust, $\tau > \pi$*, every law above*	M	M

Proof. (4) is Remark 23 with Theorem 8. Row one is Propositions 27 and 28; row two Proposition 27 and Remark 23; rows three and four Proposition 31, Theorem 33 and the flow calculation after it; rows five and six Proposition 31 and Theorem 35; row seven Proposition 31 with Corollary 25. □

Read down the table. Reading having become as being real, being unreal from b does not imply being open: rows two to four, six and seven. Being guaranteed is a lower bound that says nothing about what a law adds: rows two to four and six. The same metric gives different answers under different laws: rows three to five. Determinism does not by itself make the future fixed from a standpoint: in row one the law has unique Cauchy evolution and nothing beyond the past is fixed. That verdict belongs to the class as well as to the law. Row one is witnessed by solutions of infinite energy, and for a massive field on Minkowski spacetime the verdict changes with the class: all smooth solutions give $J^-(b)$, while in one space dimension finite-energy solutions are fixed everywhere by the past of any event (Proposition 30). On an eternal flat background, the past of an event fails to fix the future only through what can arrive along null directions, or through solutions that a finiteness condition would exclude. And an exact verdict can change while the fields barely do: by (3), the differences hidden at a massless-fixed target are $O(m^2)$ in a stated norm. Exact nonfixing and practical unpredictability are separate questions.

Part III. Determination with gravity

8 Same past, different futures

A fixed background leaves room for a reply. Perhaps gravity ties things together: perhaps coupling, conserved mass or regularity at the center would make the past of an event determine what later happens there. Subject to the inputs below, it does not.

Work with the spherically symmetric Einstein–real massless scalar system. Use double null coordinates (u, v) with $t = (u + v)/2$ and a launch event b at $t = 0$ on the center, with future unit

vector n along it; r is the area radius and m the Hawking mass. Choose incoming characteristic data at past null infinity from a compact family: fix $0 < A < T_1$, smooth F_k with disjoint supports about T_k inside $(A, 2T_N + 1)$ and $F_k(T_j) = \delta_{kj}$, and a smooth nonzero G supported after $2T_N + 1$. Take x in a small closed ball on which β below is smooth and positive. Set

$$\Phi_{\varepsilon,x} = \varepsilon\Big(\sum_k x_k F_k + \beta(x)G\Big), \qquad \beta(x) = \Big(\beta_0^2 - \sum_k x_k^2 \|F_k\|_2^2/\|G\|_2^2\Big)^{1/2}.$$

Disjoint supports make $\|\Phi_{\varepsilon,x}\|_2$ independent of x, so the total incoming mass does not depend on the target, and the tail G pays for the change after every reading.

Theorem 38 (Same past, open readings)**.** *Fix $N \geq 1$ and $0 < T_1 < \cdots < T_N$. Subject to the external inputs below, there are $\mathcal{E}_*, c > 0$ such that for every $0 < \mathcal{E} < \mathcal{E}_*$ and every $y \in \mathbb{R}^N$ with $|y| < c\sqrt{\mathcal{E}}$ there is a spherically symmetric solution (M, g, ϕ, b, n) on $\mathbb{R} \times \mathbb{R}^3$ such that:*

1. *the entire pointed causal past $J^-(b)$, with its field and launch tangent, is the same Minkowski past with $\phi = 0$ for all y;*
2. *the iterated incoming boundary mass $\lim_{V\to\infty} \lim_{u\to-\infty} m(u, V)$ equals $\mathcal{E}$;*
3. *the central future unit geodesic γ with $\gamma(0) = b$, $\gamma'(0) = n$ exists for all proper time, and $\phi(\gamma(T_j)) - \phi(b) = y_j$ for each j.*

The selected solutions have smooth representatives that are future and past causally geodesically complete, C^0-inextendible maximal globally hyperbolic developments.

Structure of the proof. External inputs. Theorem 1.1 and Definition 2.1 of Luk et al. (2016) give existence and uniqueness on finite cones. Their (1.7)–(1.9), Proposition 3.18 and Section 4 give estimates uniform in the cone vertex, and Theorem 1.7 with Section 6 gives causal completeness. Their local well-posedness proof and part of the u-side higher estimates are omitted or sketched in the source. The past half of their completeness theorem is stated in one sentence, and the extension to our compact family is our deduction from their uniform bounds. We retain all of these as inputs.

Flat wedge. On $v < A$ the regular function $p = r\phi$ satisfies $p_{uv} = Vp$ with bounded V, vanishing incoming data and a regular axis. An integral Grönwall argument gives $p = 0$, hence $m = 0$ and the Minkowski geometry there. Along a future causal curve v does not decrease, so $J^-(b)$ lies in the wedge.

Degree before the limit. On each finite cone with vertex u_n let $H_n(x)$ be the N readings. Compactness and characteristic uniqueness make H_n continuous. No uniform continuity in n and no continuous global scattering map is used. Explicit estimates give $|\varepsilon^{-1}H_n(x) - x| \leq C\varepsilon^2$ uniformly, so for small ε the Brouwer degree gives a root of $H_n(x) = \varepsilon z$ for each n.

Limits and mass. Choose a convergent subsequence of roots and pass to a global reduced solution by compactness, taking the cone limit before $u \to -\infty$. On a fixed finite interval the exact mass flux identity then has integrand tending uniformly to $\Phi^2/4$, so $\mathcal{E} = \|\Phi\|_2^2/4$. Choosing constants gives the quantifier order stated.

Regularity. A finite-order reconstruction gives a $C^{1,1}$ metric and scalar in Cartesian charts with unique geodesics. An all-order induction on the early exterior gives a smooth entire slice. A polar-areal first-order system, with a constrained coefficient that gains a derivative, propagates smoothness by localized H^3 persistence, weak–strong uniqueness and a tame continuation estimate. Polar slices are Cauchy. Causal completeness with Theorem 1 of Galloway et al. (2018) gives C^0-inextendibility, and a field-preserving common extension (Ringström, 2009) gives maximality. □

Four qualifications matter. The constants depend on the schedule, so this is not one resolution for all N; every finite sign pattern of readings occurs at N-dependent amplitudes. The fixed mass is a total: the mass arriving by each reading time is not fixed. Incoming mass is not identified with ADM mass. And the argument is target by target, so no scattering map is selected.

The readings differ because data from outside $J^-(b)$ reach the center. That a past need not fix later readings is known in general (Geroch, 1977; Manchak, 2008). That is the expected mechanism, and the conclusion should not be oversold. For the linear wave equation on Minkowski spacetime, $\phi(t,0) = \Phi(t)$ exactly, and every small central history is prescribable with the same past. That freedom is carried by a massless field arriving from past null infinity. A massive linear field of finite energy has none of it on Minkowski spacetime (Proposition 30). Whether Theorem 38 has an analogue for a massive scalar is open; the gravitational field itself remains massless. The class and its global regularity are in print (Kehrberger, 2021), and the method of finite control by disjoint bumps and a topological theorem is established (Kehle and Unger, 2022). In another category Carlotto and Schoen show that positive mass need not transmit data inward (Carlotto and Schoen, 2016). What Theorem 38 adds is an exact statement of the coupled form, with a mass and a complete maximal category held fixed.

A worry remains: perhaps these are one spacetime viewed from different marks.

Proposition 39. *Compare solutions with their central geodesic as a distinguished unparametrized curve Γ, under time-orientation-preserving field isometries preserving Γ. If the central scalar trace is C^1 and $N \geq 2$, an open ball of readings cannot be realized by countably many isometry classes.*

Proof. Such an isometry restricts to a translation of proper time on Γ. So every marked reading vector from a copy of one solution lies on the curve $\beta \mapsto (\phi(\gamma(\beta + T_j)) - \phi(\gamma(\beta)))_j$, or its negative if sign reversal is allowed. That curve is locally Lipschitz in β, so it has N-dimensional measure zero for $N \geq 2$. A countable union of null sets contains no ball. □

Bare metric nonisometry is a stronger claim and needs the center to be intrinsic.

9 Whole-realm rigidity

Agreement on $J^-(b)$ is not the only base a theory might hold fixed. Correia and Rosenkranz distinguish a pointy from a bow-tie relativistic growing block (Correia and Rosenkranz, 2018, Chap. 9). In the notation of Section 10, these are $E_b = J^-(b)$ and $E_b = M \setminus (J^+(b) \setminus \{b\})$. The bow tie contains the elsewhere, and so contains exterior radiation that the causal past misses. A past-agreement counterexample is not a counterexample to determination by that larger realm.

On a fixed background this is immediate. Put $W_b = I^-(I^+(b)) \setminus J^+(b)$. In general W_b is smaller than the bow tie, which is why the sufficiency below is the stronger claim. If two solutions of a law with the domain-of-dependence property agree on W_b, they agree on $J^+(b)$: every past-inextendible causal curve from a point of $J^+(b)$ meets an early Cauchy slice at a point of W_b. This is sufficiency, not a claim that all of W_b is needed; exact dense data already determine continuous fields, and the least region intercepting the relevant curves is not a minimal information base.

With gravity the comparison is between two spacetimes, and the theorem needs care.

Lemma 40. *In the class of Theorem 38, every two events share a chronological future. Writing $W = M \setminus J^+(b)$, every early level $\Sigma = \{t = t_0\}$ with $t_0 < 0$ is a Cauchy surface of M contained in W, and is Cauchy in W.*

Proof. In the global Cartesian coordinates of the $C^{1,1}$ reconstruction, whose causal curves are those of the smooth representative, the metric has $g_{tt} = -F$, no time–space terms, and spatial eigenvalues F and $\mathcal{A}$ with $2/9 \le F \le 8/3$ and $4/9 \le \mathcal{A} \le 1$. So $(1, v)$ is timelike for $|v| \le 1/4$, and straight paths to $(T, 0)$ with T large give a common chronological future. For the second claim, t increases along future causal curves and $t(b) = 0$, so $\Sigma \subset W$. The set W is an open past set, so a past causal curve starting in it stays in it, and an inextendible curve of W cannot end at finite $t < t_0$, since the only ambient boundary of W lies in $t \ge 0$, nor escape to spatial infinity, by the cone bound. □

Since every two events share a chronological future, $W_b = W$ in this class, and W is the bow tie minus b.

Theorem 41 (Whole-realm rigidity)**.** *Let M_1, M_2 be smooth maximal developments in this class with marks b_1, b_2. Suppose $h : W_1 \to W_2$ is an onto time-orientation-preserving isometry of metric and scalar. Then there is an onto smooth field-preserving isometry $\Psi : M_1 \to M_2$ extending h, with $\Psi(b_1) = b_2$. If h also matches the launch germ and tangent, all corresponding future proper-time readings agree.*

Proof. Let Σ_1 be as in Lemma 40 and $\Sigma_2 = h(\Sigma_1)$. Then Σ_2 is Cauchy in W_2 and acausal in M_2, since W_2 is causally convex. Every ambient inextendible causal curve has a nonempty past tail in W_2, and its maximal connected portion there is inextendible in W_2, so it meets Σ_2. Hence Σ_2 is Cauchy in M_2. The isometry matches the induced metric, extrinsic curvature, scalar and its normal derivative on these surfaces. Both spacetimes are maximal developments of those data, so uniqueness (Choquet-Bruhat and Geroch, 1969; Ringström, 2009), in the scalar form of Ringström (n.d.), gives an onto smooth field-preserving isometry Ψ with $\Psi|_{\Sigma_1} = h|_{\Sigma_1}$. Ψ and h have the same first jet there, so they agree on the connected set W_1, and Ψ carries $J^+(b_1) = M_1 \setminus W_1$ onto $J^+(b_2)$. Thus $J^+(\Psi(b_1)) = J^+(b_2)$, and $\Psi(b_1) = b_2$ since M_2 is causal. The last claim follows from uniqueness of geodesics. □

In one physical class, with one deterministic law:

Held fixed	Future central readings
Whole pointed causal past, and total incoming mass	Open ball
Whole open realm W, with marked germ	Fixed, and so is the whole development

The word "present" cannot be substituted for both bases. Neither result says which, if either, is what exists from b, and the second does not make the determined future exist. It does show that a theorist of the bow-tie realm has a future fixed by what exists from b, while a theorist of the pointy realm in the same class does not. That holds for the bow tie generally, since it contains a Cauchy slice; the theorem's added content is the transport between two spacetimes. Openness relative to a realm depends on the realm chosen and the law, and the choice of a realm is not forced by physics.

10 Realm structure

Which realms are available? Write E_b for a point-level assignment over a fixed causal frame. This is a metalinguistic device, and implies no claim that the frame is concrete. Two conditions do much of the work: $b \in E_b$, and persistence along the causal order, $b \le c \Rightarrow E_b \subseteq E_c$.

Proposition 42. *Self-inclusion and causal persistence imply $J^-(b) \subseteq E_b$. With future exclusion $E_b \cap J^+(b) = \{b\}$ and no elsewhere point this gives $E_b = J^-(b)$; with every elsewhere point it gives the bow tie.*

Proof. If $x \leq b$ then $x \in E_x \subseteq E_b$. The rest is the partition of M into past, future and elsewhere. □

No field equation, Lorentz invariance or transitivity enters. The persistence condition matters: with persistence only along timelike order, $I^-(b)$ is available too. Correia and Rosenkranz's clauses use the full causal order (Correia and Rosenkranz, 2018, pp. 144, 148–149). A presentist denies persistence. Choosing between pointy and bow-tie needs further premises. Under the locality premise of Part I no intermediate choice is available in flat spacetime (Corollary 12). Requiring existence realms to be closed would favour the pointy profile, but metric regularity does not supply that; it is a bridge. The pair itself is not new. Bigaj (2008) obtains the same two relations from Stein's requirements in Minkowski spacetime, with an exhaustiveness stipulation and cumulativity in place of transitivity. Proposition 42 reaches them from different premises, and neither result implies the other.

The bow tie is also not always what a natural criterion gives. Let $D(b)$ consist of b and the points outside its strict causal future that share a strict causal successor with it. On connected causally simple spacetimes this is persistent exactly when every pair of events has a common strict causal future. In the class of Theorem 38, Lemma 40 gives $D(b)$ equal to the bow tie. Elsewhere it need not be: on a complete conformal cylinder of duration L over a complete spatial manifold S, the least persistent enlargement of D equals the bow tie at every standpoint exactly when $\operatorname{diam}(S) \leq L$. Whether a horizon obstructs the criterion depends on the class.

Part IV. Existence and truth

11 Stages, ended worlds and maximality

A truncated spacetime can represent two different things: what exists so far, on a growing view, or a whole world that stops. Conflating them makes an openness claim an artefact of representation.

Let (M, g, ϕ) be a smooth member of the class above with global center-normalized time T, and fix $T_0 < T_*$. Then $M_< = \{T < T_*\}$ is an open smooth solution with the same complete Cauchy data on $\{T = T_0\}$, whose constant-T slices remain Cauchy, and for T_* after b the pointed past of b is unchanged. The field equations fail nowhere on it. So three principles have different strengths:

Local lawfulness: every included point satisfies the equations. Both M and $M_<$ do.

Continuation is possible: a regular end with no boundary obstruction admits a lawful extension.

Maximal-world admissibility: complete histories representing physically possible worlds are maximal developments, unless a physical end rule is stated.

Only the third excludes $M_<$ as a complete world, and it is not a consequence of the first. It is a defensible methodological choice for comparisons that claim to hold a theory fixed: choosing a shorter domain adds a contingent restriction with no boundary mechanism. It is not a refutation of brute endings.

Accepting maximality concedes no ontology. A growing theorist can use M to represent everything that does, did or will happen while assigning a proper inventory to each standpoint. So growth with a physically determined future is consistent at this level, which refutes any inference from growth alone to indeterminism. A completed mathematical representation and an ontological completed history are different commitments. The causal-set case makes the same point sharply: the prescribed complex percolation values extend to a countably additive measure on the labelled doubly-principal cylinder algebra exactly when $p \in [0, 1]$ (Impellitteri, 2026a). A failure of completion in a growth theory is a mathematical fact about a measure. It neither refutes becoming nor, where it succeeds, establishes a block.

12 Truth

Fix a claim q at b about a reading at a later event. Four claims should be separated. *Outcome existence*: in each admitted complete realization the target occurs with a definite value. *Settlement*: every realization admitted by the chosen base gives q the same value. *Truth now*: q has an absolute truth value at b. *Grounding*: an appropriate fact explains that truth, perhaps under a later temporal operator.

Take two realizations sharing a stage at b, with the target absent from the inventory at b and included when it occurs, and with the value differing between them. On policy L, evaluated in a designated realization, q is bivalent at b, and its truth can be assigned a later ground. This is the restricted form of the later-grounded view (Correia and Rosenkranz, 2018), which Wawer (2021) challenges. On policy S, q is evaluated over both realizations and is neither supertrue nor superfalse at b. Both policies have eventual occurrence, a definite eventual value and the same change of inventory. They disagree about truth now. So eventual resolution with growth does not entail truth now, and it does not exclude it. This is a restricted counterinterpretation, not a model of any named theory or a refutation of bivalence, and the logical point belongs to the debate over future contingents. Policy S is supervaluation over the admitted realizations. Policy L evaluates in a designated realization, as thin-red-line semantics does, and it inherits the question of what designates it. A third option, on which the truth of an utterance is relative to a later context of assessment (MacFarlane, 2003), is not considered here.

The physics fixes the comparison and not the policy. Theorem 38 supplies realizations sharing a pointed past. Which of them count as admitted, and whether one is designated, is the policy. Designating an actual realization is not derived from the past of b.

13 The direction of time

Nothing so far says why the past should be the base. Write $F^-_{\mathcal{L}}(b) = F_{\mathcal{L}}(I^-_{\mathrm{s}}(b))$ and $F^+_{\mathcal{L}}(b) = F_{\mathcal{L}}(I^+_{\mathrm{s}}(b))$, and let $\mathrm{Det}^+(b)$ be the set of events all of whose future-inextendible causal curves meet $I^+_{\mathrm{s}}(b)$.

Proposition 43 (Mirror)**.** *Let θ be an isometry of M reversing time orientation, and let $\mathcal{L}$ be invariant under pullback by θ. Then $\theta(F^-_{\mathcal{L}}(b)) = F^+_{\mathcal{L}}(\theta(b))$ and $\theta(\mathrm{Det}(b)) = \mathrm{Det}^+(\theta(b))$. Without such an isometry, every result of Parts II and III that is stated for a time-reversal invariant law holds for the same spacetime with its time orientation reversed.*

Proof. A difference of solutions vanishing on $I^-_{\mathrm{s}}(b)$ pulls back to one vanishing on $I^+_{\mathrm{s}}(\theta(b))$, and θ exchanges past- and future-inextendible causal curves. The last claim holds because those results use the orientation only as a label. □

The proposition is elementary, and its consequences are not idle. In Minkowski spacetime the conformal scalar is fixed from b's strict future exactly on $J^+(b)$. In the closed dust model the reflection $\eta \mapsto 2\pi - \eta$ is such an isometry, so $\mathrm{Det}^+(b) = M$ for $\eta_b < \pi$. The statement that the guarantee changes with cosmic epoch is true, and it has an exact mirror. The Einstein–scalar system is time-reversal invariant, so Theorem 38 has a twin in which a causal future and total outgoing mass leave earlier central readings free. Theorem 8 carries its orientation in (CB); reverse it and the same proof gives the future lobe.

The mirror relates a past base at b to a future base at $\theta(b)$. It does not compare the two bases at one standpoint. For that comparison the right objects are not the fixed events but the fixed quantities.

Proposition 44 (Reciprocity and balance)**.** *Let $\mathcal{L}$ be a vector space V of real fields on M, or a translate of one, and for $A \subseteq M$ let $K_A = \{\varphi \in V : \varphi|_A = 0\}$. Let $A, C \subseteq M$.*

1. *A quantity that depends only on the field on C is fixed by the field on A if and only if it is a function on $V/(K_A + K_C)$. So what A fixes about C and what C fixes about A are the same quantities.*

2. *The freedom left on C when the field on A is given is $K_A/(K_A \cap K_C)$, and the freedom left on A when the field on C is given is $K_C/(K_A \cap K_C)$.*

3. *When these spaces are finite-dimensional, their dimensions differ by* $\dim V|_C - \dim V|_A$.

Proof. A function on V depends only on the field on A exactly when it is constant on cosets of K_A, and a function constant on cosets of K_A and of K_C is constant on cosets of $K_A + K_C$. The fields on C compatible with a given field on A form a coset of the image of K_A under restriction to C, and the kernel of that restriction is $K_A \cap K_C$. The last claim follows from $\dim V|_A = \dim V - \dim K_A$. □

In particular no reading fixes another one way. If $y, z \notin F_{\mathcal{L}}(S)$ and $y \in F_{\mathcal{L}}(S \cup \{z\})$, then on K_S the evaluation at y is not zero and vanishes on the kernel of the evaluation at z, so it is a nonzero multiple of it, and $z \in F_{\mathcal{L}}(S \cup \{y\})$.

For deterministic dynamics that preserves phase volume, Albert already observes that whatever constrains the world's condition at one time constrains it at every other "to exactly the same degree", and that the constraints differ in form and not in content (Albert, 2000, pp. 121–122, n. 11). Proposition 44 is the analogue for two regions and a linear class, where the two sides need not carry the same amount. So two bases can differ in what each leaves open about the other, and only by the difference in what they carry. No symmetry of the law is used, and the statement holds for linear laws that are not reversal invariant. An event of C lies in $F_{\mathcal{L}}(A)$ when its reading is one of the shared quantities; the sets $F_{\mathcal{L}}(A) \cap C$ and $F_{\mathcal{L}}(C) \cap A$ need not correspond, because a shared quantity can be a point reading on one side and not on the other.

Corollary 45 (An arrow from epoch)**.** *On the closed dust model let $m > 0$ and $b = (\tau, Y)$. Then $F_m^+(b) = M$ for $0 < \tau \leq \pi$ and $F_m^+(b) = J^+(b)$ for $\pi < \tau < 2\pi$. Consequently, for $\tau < \pi$ the field on $I_{\mathrm{s}}^+(b)$ fixes the whole solution while the field on $I_{\mathrm{s}}^-(b)$ does not; for $\tau > \pi$ the reverse holds; and for $\tau = \pi$ each fixes the whole solution.*

Proof. The reflection $\eta \mapsto 2\pi - \eta$ is an isometry reversing time orientation, and $\mathcal{L}_m$ is invariant under pullback by any isometry. Apply Proposition 43 to Theorem 35. □

In the terms of Proposition 44, with $A = I_s^-(b)$ and $C = I_s^+(b)$: before maximum expansion $K_C = 0$ and $K_A \neq 0$, so everything about the past of b is fixed by its future and its future is open given its past. This is one law, one model and one standpoint, with the two bases on an equal footing. The law is reflection invariant and so is the geometry. The asymmetry at b comes from where b sits: the future of an early event contains a Cauchy surface and its past does not.

So the laws give exact fixing no arrow of their own. A model can still have one, as Corollary 45 shows, and Proposition 44 says what it consists in: an imbalance in what the two bases carry. The arrow of Corollary 45 points toward the larger side of the universe and reverses at maximum expansion, so it is not the asymmetry of ordinary thought. "The past is fixed and the future open" cannot be read off the laws, and where global geometry does not supply it the asymmetry needs another source. We describe three candidates and are neutral between them. They are not the only ones. Each gives (CB) a different meaning.

Thermodynamic. The dynamics is reversal invariant, and statistics over a present macrocondition make higher entropy overwhelmingly probable on both temporal sides. So Albert adds a law-like posit, the Past-Hypothesis, that the world began in a particular low-entropy macrocondition (Albert, 2000, p. 96). Prediction uses the dynamics and the statistical rule, while most of what we know of the past rests on records, and that asymmetry is meant to follow from the posit (Albert, 2000, pp. 113–116). On this view the past is a natural base because it is what we have records of, and that is explained by a boundary condition and not by $\mathcal{L}$.

Perspectival. Price warns against accepting an argument in one temporal direction while rejecting its mirror (Price, 1996, chs. 1–4). He separates the asymmetries of dependence, causation and agency (Price, 1996, pp. 132–133) and defends a perspectival view of causal asymmetry. He presupposes that the present and flow are perspectival, so he is a party to the dispute. Proposition 43 is the fixing instance of his warning. His treatment of cosmology also bears on Part II: if a low-entropy boundary needs explaining at one end, a time-symmetric Gold universe cannot be dismissed by arguments that assume the arrow (Price, 1996, p. 81ff).

Causal. Tooley takes causal direction as basic and bases temporal direction on it (Tooley, 1997). A cause is actual as of its effect's time and not conversely (Tooley, 2010). On this view the asymmetry of openness would be ontological. He also defends absolute simultaneity, which is the clock-equipped horn of Section 3.7.

Others. Further accounts place the asymmetry in the agent and not in the inventory. Torre (2011) surveys two: that the future depends counterfactually on the present as the past does not, and that we have power over the future and not the past. Ismael (2012) takes the open future to be how time appears from the standpoint of a deliberating agent, and Kutach (2011) takes the fixity of the past to be an asymmetry of useful influence. None makes the asymmetry a consequence of the laws alone, which agrees with Proposition 43.

One question stays open. If $\mathcal{L}$ includes the Past-Hypothesis as a condition at one temporal end, it is not reversal invariant, and Proposition 43 fails. A selection of solutions can create one-way exact fixing only by making one reading a non-injective function of another, which the remark after Proposition 44 excludes for linear classes and a selection can restore. We expect a macrocondition to do this for few exact readings, since it does not fix exact values, and the asymmetry to appear in conditional probabilities over completions. There it has a definite size. For discrete readings a of the past base and c of the future base, under any probability measure on $\mathcal{L}$, $H(c \mid a) - H(a \mid c) = H(c) - H(a)$: the future is more open than the past, each given the other, by the amount by which its description carries more entropy. If so, chance given a base belongs beside exact fixing, and much ordinary talk of openness may mean that. We have not proved it.

14 The upper bound and the lower bound

14.1 What each side owes

Let R_b be what a restricted view counts as concrete from b, and H the represented history. Constitution gives $R \subseteq E_b$ for an activity region R, which may include events later than b (Section 15). Persistence gives $J^-(b) \subseteq E_b$ (Proposition 42). A restricted view adds $R_b \subseteq E_b$. These are lower bounds, and every argument from physical organization that we have examined stops at one. Theorem 8 gives an upper bound only by amplifying an assumed failure of universality, as discussed below. A lower bound cannot settle the matter, because $R_b \subseteq E_b$ is compatible with both $E_b = R_b$ and $E_b = H$.

	Pointy restricted	Slice or bow-tie restricted	Event realist
$E_b \subseteq H$	Holds for every completion through b	Only for completions agreeing on the slice	Holds of one H
Selection owed	None	For that part	Which completion, where they differ
Extra claim	$E_b \subseteq R_b$, negative, differs by b	The same	$H \subseteq E_b$, positive, the same for every b
Must say	Whether more will come, or why that is unsettled	The same	Why H and not a truncation
Supported by $R_b \subseteq E_b$?	No	No	No

So the two debts are not of one kind. Where H is unique in a class all parties accept, the event realist's extra claim is a stronger lower bound. Where completions are not unique, the event realist's $E_b \subseteq H$ is a selection that a pointy theorist does not owe. Three views fall outside the table: on McCall's view as reported by Earman (2008) E_b lies in no single history, fragmentalists have no one H, and for a theorist whose H has no referent beyond R_b, "nothing concrete lies outside H" is the thesis itself. The positive claim carries more events and the negative one a changing limit. We know of no neutral way to price that exchange.

14.2 Totality

Is "that's all" an instance of a known problem? Armstrong's totality fact relates a mereological whole to a property when the whole comprises all things with the property, and serves as truthmaker for negative and general truths (Armstrong, 2004, §6.2). The property may be abundant and time-indexed, and he grants that the relation sets a limit (Armstrong, 2004, pp. 72–73). Molnar argues it is a negative fact under another name (Molnar, 2000). The rival accounts differ only about the truthmaker: absences, the world with its contents essential (Cameron, 2008), the whole positive fact (Cheyne and Pigden, 2006), or no truthmaker, on optimalist views with Bigelow's supervenience of truth on being (both as reported in MacBride, 2024). All start from an accepted negative truth and ask what grounds it. None is a reason to accept it. Filled as $T(R_b, \text{concrete from } b)$, the slot restates the claim; if the property is membership in R_b, it is trivial. Forrest's use of general facts in a growing block (Forrest, 2006) appears to do other work, but we have not read it beyond its abstract and one review, and it could change the first row of the table below.

The neighbouring literature has the core points. Earman says a growing-block theorist cannot deny the future absolutely without fixing one truncated model and giving up growth, so the denial is perspectival, and a block theorist can say the perspectival version as a triviality (Earman, 2008, pp. 144–145). He separates this rigid denial from a shifting one, that far enough beyond now nothing exists, which holds at every stage and fails in block models (Earman, 2008, pp. 144, 159). His "absolute" is not Fine's: the standard realist holds a tensed denial absolutely and keeps growth through what will be, and Deng rejects the step from that view to one fixed block (Deng, 2017, §4). Deng also objects to Tooley that what is actual as of now includes that those facts are all the tenseless facts, and that this is tensed (Deng, 2017, §6). None of these sources discusses the inference from constitution to inventory, so they do not state the gap above. No priority audit has been run.

14.3 Persistence and the rigid claim

Persisting truths are closed under entailment, so a claim that fails to persist has no persisting entailers. "Everything lies in R_b", with R_b rigid, fails to persist wherever the inventory changes, on a growing block or on presentism. That is the reason behind Earman's rejection of a rigid denial. It does not touch the shifting form "everything is past or present". The rigid instance follows from the shifting form plus self-location, so the only non-persisting ingredient is locational, and the debt moves to the shifting form with an unrestricted quantifier. The observation covers positive tenseless and past-tensed premises. It leaves open past-tensed conclusions and non-deductive support. It does not cover totality-form constitution claims or a presentist's present-tensed premises. It is close to the familiar point that tenseless premises do not yield tensed conclusions, and it says nothing against the event realist.

Proposition 46. *A growing-block theory cannot hold truthmaker maximalism, temporal necessitation, permanence of what exists, and a rigid "that's all" truth at b which growth falsifies.*

Proof. This is Loss's argument (Loss, 2022, §II), with growth in place of A-change. If p is true at b it has a truthmaker, which by permanence exists at a later stage, and by necessitation p holds there, contrary to growth. □

Given the shifting form, the rigid claim is equivalent to "b is present", which is Loss's own case, and his §V treats presentness on the block as being at its edge. Presentism lacks permanence and the block view lacks change, so neither is touched. The exits are these.

Member given up	Cost
Permanence	A fact ceases at each step, against Broad's text and Loss's (GB2)
Temporal necessitation	Truthmakers gain proper parts, the block is the truthmaker and mereological universalism goes (Loss, 2022, §§V–VI). Molnar's rejection of contingent truthmaking (2000, §IX) targets a primitive relation; whether it reaches this version is unargued
Maximalism	Growth stays additive, and the upper bound is no fact at the fundamental level. Cameron and Cheyne and Pigden argue against the restriction
Truth-value change	Index the proposition, as Tooley does (1997). Actuality as of a time is primitive, and Deng's objection applies

Separately, each view takes a stance on Fine's four postulates (Fine, 2005), as Deng sets them out. Those stances are not exits from this table: relativizing every clause to standpoints reproduces the contradiction at the later standpoint, and a standard realist faces it with tense operators. Nothing above needs a foliation, since a fact ceases at s if it holds at s and fails at every later stage, which needs only a partial order. What relativity removes is a privileged chain of stages, and without one, parity pushes a view towards relativism. That would hold in Newtonian spacetime too.

14.4 Stated candidates

The following table sorts the stated upper-bound candidates we know. It covers these candidates only; it is not a theorem about all such principles.

Candidate	Index	Outcome
Armstrong totality	None	Restates the claim
Maximal-world admissibility	None	Bounds a complete history, not E_b
Theorem 8 via the reality bridge	Defined or primitive	Conclusion shared if defined; non-universality rejected if primitive
Interaction closure (Brading, 2015)	None	Inclusive as stated; with its converse it falls at H if connected
Causal criterion, tenseless (Colyvan, 1998)	None	Later events pass
Causal criterion, indexed	Defined or primitive	The same bound as Theorem 8, given the bridge and non-universality
Tooley: an effect is not actual as of its cause's time	Primitive	Rival rejects. Support comes from an analysis of causation, and the direction it needs is disputed (Section 13). Paired with absolute simultaneity
No fact selects a completion, the reply Earman reports (Earman, 2008, pp. 144, 154, 159)	Primitive	Rival rejects the bridge. Silent in deterministic classes
Least truthmaker base, given settled truth	Primitive	Abductive. Rival rejects the truth policy. Our construction

Theorem 8 is a real upper-bound derivation, of a conditional kind. It amplifies an assumed failure of universality into a sharp bound and does not supply the first negative. The bound is on what has happened by b, and an eternalist can accept that. Existence enters only at a bridge, which reads having become as being real.

The causal criterion deserves a remark, because it is the closest thing to a sourced upper bound. The Eleatic principle admits only what has causal power. Read tenselessly it passes every later event with causes or effects, so it gives no restriction at the future. Only a standpoint index does: "concrete from b only if it can affect b". That is the converse of Clifton and Hogarth's maxim, and the maxim's rationale, that an unreal event could not help bring b about, gives no support to the converse. Colyvan notes that the epistemic motivation for the principle gives no reason to believe in future objects, and counts a similar exclusion of distant regions against that motivation (Colyvan, 1998).

The diagnostic. An index on "concrete from b" can be defined or primitive. Defined as a causal or temporal restriction, the claim is a triviality every party accepts. Primitive, it is a thesis the event realist rejects, indexed or not.

So "indexed" sorts nothing. Correia and Rosenkranz's clause (P2R) and Tooley's actuality as of a time are indexed and rejected. Static views give no sense to indexed actuality (Tooley, 2010). For Fine's relativist, what is said from b still carries the index, so "absolute" is not available as a contrast. The missing argument lies between the defined and primitive readings. Three of the entries above, Tooley's, the no-fact principle and the least-base argument, offer support of a different kind, and they fall outside this sorting.

15 Constitution and process

One physical route to a restricted inventory runs through systems. If a genuine unity is constituted by an activity extending through a region R (Brading, 2015), perhaps R is what exists for that system. The inference fails at one identifiable step. Grant that the unity's realizing occurrences must be concrete. Compare $E_b = R$ with $E_b = H$. Both satisfy $R \subseteq E_b$, so an argument whose only inventory requirement is that constituting occurrences are concrete cannot distinguish them. It supplies a lower bound. A process can have exactly its own parts while other events are also concrete. So the route from constitution to an exhaustive inventory is closed as a deductive route unless a further exhaustiveness premise is supplied, and adding examples will not supply it. The revision that makes the activity the object's parts earns a process mereology, and the rival can accept that.

An explanatory route remains: does a growing inventory explain process unity better? Take a feedback-controlled filling device with a sensor, a stored target and a valve, and compare a timed open-loop replay that closes at the same moment. For one supply rate the two have identical liquid height and flow histories. Under a changed rate the feedback device reaches its target later, and the replay stops short. So organized causal roles explain something a list of outputs does not, and a process is not a rigid aggregate of its actual events. But an event realist who identifies the operation with the causally organized execution, with an explicit participation rule, keeps all of that, and so does a growing theorist. The argument for an exclusive advantage assumed that event realism must identify a process with a rigid aggregate, and it need not (Stout, 1997; Steward, 2013). This is a comparison for one explanandum, not a verdict of parity, and whether a further form or grounding fact is needed remains open.

16 Which inferences fail

The results license no ontology, but they do bear on particular inferences. None of what follows is claimed as new.

Metric-only shared presents. Corollary 9. A transitive relation containing the chronological past and defined from the metric alone has no shared present if it is local. Its defenders must name the premise they reject: locality on GH, transitivity, the off-shell premise, or the category.

The causal past as a base for every rival. Theorem 41. A comparison must match the rival's realm. A past-agreement counterexample does not reach a bow-tie view, whose realm contains a Cauchy slice.

Causal limits on prediction read as openness. Theorem 37. What is guaranteed does not bound what a law fixes, as Manchak (2008) and Earman (1995) partly anticipate.

"Past fixed, future open" as evidence for growth. Proposition 43. The slogan cannot be read off the fixing relation. The asymmetry of records has candidate explanations that use no inventory distinction (Albert, 2000; Ismael, 2012; Kutach, 2011), so it is not evidence for growth until they are shown to fail. The growing theorist can reply that it does fail. This removes one support and refutes nothing.

Growing-block maximalists. Proposition 46. This is Loss's argument, and it fires through the rigid claim.

Unreality to law-relative openness. Theorem 37, on the reading of having become as being real. The distinction is old; the models are exact. Openness of existence is untouched, since it concerns the modal class (Section 11).

Past-relative nonfixing to growth. Theorem 38 holds for an eternalist, since each model is one whole spacetime with its own exterior data. It is no evidence against indeterminist growth, and we know of no one who argues this way from it.

Reality to truth. Putnam (1967, pp. 243–246) argues that future contingents already have truth values. That step needs a policy connecting outcomes to earlier truth (Section 12). A block theorist can adopt the policy at no cost to the inventory, so the reason is to state it.

Constitution to inventory. Section 15, against the argument as we reconstructed it.

Event realism as a default. Where completions are not unique, the selection of H is a debt. A selection principle preferring fewer primitives (Impellitteri, 2026b) needs a defence of its own.

One reading of Stein (1991, pp. 148–149) is also excluded. If "definite" and "unsettled" meant fixed by law, the relation would not be J^-, since Theorem 37 has $F = M$ in several rows with B inside the causal past. Nothing we have read shows that Stein intends that reading, and his theorem does not need it.

The positions compare as follows. Each row is a selected profile, not a reconstruction of a whole theory.

View	Concrete from b	Premise denied or added	Future fixed from what exists?
Presentism	Present only, however spatially read	Denies persistence; owes an upper bound	Depends on the spatial reading and the law: as the clock row for a Cauchy slice; past-grounding owed
Pointy growing block	$J^-(b)$	Owes an upper bound	Open in Theorem 38; fixed in closed dust after maximum expansion
Bow-tie growing block	Past and elsewhere	Rejects transitivity of existence-at; owes an upper bound	Fixed wherever the law has unique evolution, since the realm contains a Cauchy slice
Standpoint relativism	$J^-(b)$, relative to b	Rejects Absolutism	As pointy; the geometry is agreement among standpoints
Clock-equipped view	$\{T \leq T(b)\}$, with a level of T as present	Changes the category	Fixed if the law has unique Cauchy evolution
Event realism	H	Owes a lower bound, and selection where completions are not unique	Yes, trivially, and that says nothing about law

17 What would change these conclusions

A gluing counterexample, or a category mismatch in Part I. A proof that on-shell energy conditions change the classification; the pocket problem of Section 3.6 is the route. A failure of the external inputs of Theorem 38. A prior theorem with the same hypotheses, which would change credit and not content. In the philosophy: an independently defended tensed premise yielding the upper bound; an explanandum for which a growing inventory does exclusive work; or an argument that the defined/primitive contrast is unavailable, as a deflationist about the debate may hold (Savitt, 2006; Dorato, 2006). A successful deflation would remove the existence question and leave Parts I to III standing.

18 Conclusion

The future can be unreal and fixed, real and unfixed from any standpoint, fixed at its end before its middle, and open or closed in one physical class according to what is held fixed. General relativity with a locality premise says sharply what can have become. Laws and comparisons say what is fixed, and in the models treated we can say it exactly. Neither says what exists, and neither says what is true. What the past can fix turns out to be a question physics can answer, and it is a different question from the one it is usually taken to answer.

Note on method. This manuscript was developed with substantial assistance from large language models (Anthropic Claude, OpenAI Codex and GLM models), including literature search, proof drafting and cross-model proof checks that found and repaired errors. Review by models is internal review. It is not expert or journal review, and agreement between models is not evidence of correctness. Responsibility for errors is the author's.

A Proofs for Part I

A.1 Theorem 14, Theorem 15, Proposition 16

We use: I^+_{s} is open in $M \times M$ (MS Prop. 2.16); $\overline{J^+} = \overline{I^+_{\mathrm{s}}}$ and $\operatorname{Int} J^+ = I^+_{\mathrm{s}}$ (MS Prop. 2.17), and likewise for the pasts; if $p \ll r \leq q$ or $p \leq r \ll q$ then $p \ll q$, and if $r \in E^+(p)$, $q \in E^+(r)$ then $q \in E^+(p)$ or $p \ll q$ (MS Prop. 2.31); two events joined by a causal curve that is not a maximizing lightlike curve are joined by a timelike curve, and a maximizing lightlike curve is a lightlike geodesic (MS Thm. 2.30(i), (ii)); so a causal curve between horismotically related events is a lightlike geodesic.

Proof of Theorem 14. By Theorem 8(a), $B = I^- \cup N$. Timelike and causal curves between points of a causally convex set stay in it, so I^-, J^- and E^- are locally covariant and N inherits (LC); (NC) is transitivity restricted to horismotic pairs. Conversely, $I^- \cup N$ is reflexive and satisfies (CB); for transitivity, a composition with a chronological step is chronological by MS 2.31, and a composition of two N-steps is chronological or horismotic, in which case (NC) applies. The last claim is Theorem 7. □

Proof of Theorem 15. (NC): if $x \to y \to b$ along lightlike segments γ_1, γ_2 and $x \in E^-(b)$, then $\gamma_1\gamma_2$ is a causal curve between horismotic points, hence an unbroken lightlike pregeodesic (MS Thm. 2.30), and it contains the Weyl point of γ_1. (LC): lightlike geodesics between points of a causally convex image stay in it, and isometries preserve the Weyl tensor. On $\mathbb{M}$ the Weyl tensor vanishes. If it vanishes nowhere, every horismotic pair is in N. If M is flat on an open causally convex V and the Weyl tensor is nonzero at some point z, horismotic pairs inside V are not in N (every causal curve between them stays in V), while a horismotic pair joined by a lightlike segment through a neighbourhood of z is. The Weyl tensor, as a $(1,3)$ tensor, is invariant under conformal rescaling (standard), and unparametrized lightlike pregeodesics are conformally invariant (MS §2.6). The last case is not vacuous: the spacetime M' of Section A, built from a g with nonzero Weyl tensor at x, is flat on the open causally convex set $\{t < -3r\}$. □

Proof of Proposition 16. If B_M is closed and contains I^-, it contains $\overline{I^-} = \overline{I^-_{\mathrm{s}}} = \overline{J^-} \supseteq J^-$, so $B = J^-$ by Theorem 8(a). If $B_M \setminus \Delta$ is open and contained in J^-, it lies in $\operatorname{Int} J^- = I^-_{\mathrm{s}}$, so $B_M \subseteq I^-_{\mathrm{s}} \cup \Delta = I^-$, and (CB) gives $B = I^-$. □

Proof of Corollary 13. By worldline becoming, $b \in B_{M,\lambda'}(b)$ for every $\lambda' \ni b$; with $c = b$, worldline transitivity gives $B_{M,\lambda}(b) \subseteq B_{M,\lambda'}(b)$, and by symmetry they are equal. Call the common set $B_M(b)$. Transitivity: if $a \in B_M(b)$ and $b \in B_M(c)$, choose $\lambda \ni b$, $\lambda' \ni c$ and apply worldline transitivity. (CB): if $a \ll b$, a timelike curve from a to b extends to an inextendible timelike curve λ, and $a \in I^-(b) \cap \lambda \subseteq B_M(b)$. (LC): for $\psi : N \to M$ take $\lambda_N \ni b$ and an inextendible extension λ of $\psi \circ \lambda_N$; then $a \in B_N(b) \iff a \in B_{N,\lambda_N}(b) \iff \psi(a) \in B_{M,\lambda}(\psi(b)) \iff \psi(a) \in B_M(\psi(b))$. □

A.2 Theorem 8

The idea: a forbidden pair in a curved spacetime is moved, by local covariance, into an auxiliary spacetime that agrees with the original near the pair and is flat elsewhere; chronological becoming and transitivity then produce a forbidden pair in a flat region, contradicting Theorem 7.

Lemma 47 (Future pairs reduce to spacelike pairs)**.** *If $x \in J^+(b) \setminus \{b\}$, there is $e \ll x$ spacelike to b.*

Proof. Let U be a convex normal neighbourhood of b and $V \subseteq U$ an open neighbourhood of b that is causally convex in M (M is strongly causal; MS Remark 2.12 and the construction following it). Causal relations in U, regarded as a spacetime, are read off $\exp^{-1}$ (MS Prop. 2.10), and for points of V they agree with those of M (MS Remark 2.12). A future causal curve from b to x is nonconstant (M is causal), so it contains some $b_1 \in V \setminus \{b\}$; then $b \le b_1 \le x$ and $u = \exp_b^{-1}(b_1)$ is a nonzero future causal vector. Choose a spacelike $a \in T_bM$ with $g(u,a) > 0$ and put $q_\varepsilon = \exp_b(\varepsilon a)$. Synge's world function σ is smooth on $U \times U$ with $\nabla_q \sigma(q, b_1)|_{q=b} = -u$, so $2\sigma(q_\varepsilon, b_1) = g(u,u) - 2\varepsilon g(u,a) + O(\varepsilon^2) < 0$ for small $\varepsilon > 0$. The vector $\exp_{q_\varepsilon}^{-1}(b_1)$ depends continuously on ε and tends to $u \neq 0$, so for small ε it is future-directed timelike: $q_\varepsilon \ll b_1$. Also $2\sigma(q_\varepsilon, b) = \varepsilon^2 g(a,a) > 0$, so q_ε is spacelike to b. By MS Prop. 2.31(i), $e = q_\varepsilon \ll x$. □

The same local argument shows that every point has spacelike-separated points arbitrarily close to it.

Lemma 48 (Spacelike chains)**.** *In connected globally hyperbolic M, any two points are joined by a finite chain of consecutively spacelike-separated points.*

Proof. Globally hyperbolic spacetimes (MS Def. 3.70) are causally simple (MS Prop. 3.71), so $J^+(z)$ and $J^-(z)$ are closed and the set of points spacelike to z is open. Fix p and let S be the set of points reachable from p. S is open: if $q \in S$, pick z spacelike to q; a neighbourhood of q is spacelike to z and reachable via q, z. $M \setminus S$ is open: if $q \notin S$ and c is spacelike to q, a neighbourhood W of q is spacelike to c; were some $q'' \in W$ in S, then $c \in S$ and $q \in S$. M is connected. □

The example $M = \mathbb{R} \times S^3$, $b = (0, p)$, $x = (3\pi, p)$ shows that a single point spacelike to two causally related points need not exist; the chain is needed.

Set-up. Fix spacelike-separated $x, b \in M$. Since $J^+ \subseteq M \times M$ is closed (MS Prop. 3.68(iii), 3.71), there are small closed spacelike embedded 3-discs $D_x \ni x$, $D_b \ni b$, each acausal, with no causal curve between them; so $D_x \cup D_b$ is a compact acausal spacelike submanifold with boundary, and Bernal and Sánchez give a smooth spacelike Cauchy hypersurface $S \supseteq D_x \cup D_b$ (Bernal and Sánchez, 2006, Thm. 1.1, Rem. 4.14). Choose $\delta_0 > 0$ and a smooth embedded curve $c : [-\delta_0, L + \delta_0] \to S$, unit speed for the induced metric h, with $c(0) = x$, $c(L) = b$.

Lemma 49 (Adapted coordinates)**.** *Let n be the future unit normal to S, $(e_1 = c', e_2, e_3)$ an h-orthonormal frame along c with (n, e_1, e_2, e_3) positively oriented, $\sigma(s,y) = \exp^h_{c(s)}(y_2 e_2 + y_3 e_3)$, and $\Phi(t,s,y) = \exp^g_{\sigma(s,y)}(t\,n)$. Write $T_\rho = \{|t| < \rho,\ -\rho < s < L + \rho,\ |y| < \rho\}$. For some $\rho_0 \le \delta_0$, Φ is a diffeomorphism from $\overline{T_{\rho_0}}$ onto a closed neighbourhood of $c([0,L])$, orientation- and time-orientation-preserving; $\Phi(\{t = 0\}) \subseteq S$ while $\Phi(t, \cdot) \in I^+(S)$ for $t > 0$ and $\in I^-(S)$ for $t < 0$; $\Phi^* g = -dt^2 + h_t(s,y)$; and $\|\Phi^* g - \eta\| \le K(|t| + |y|)$ on $\overline{T_{\rho_0}}$.*

Proof. Gaussian normal coordinates to S composed with Riemannian normal coordinates of S along the compact embedded arc; points $\Phi(t,\cdot)$ lie on the normal timelike geodesic through a point of the acausal S. At $(t,y)=(0,0)$ the metric is exactly η because the frame is orthonormal on c; the estimate is smoothness on a compact set. □

Lemma 50 (Cone control)**.** *Let $\eta_c = -c^2dt^2 + ds^2 + |dy|^2$, $\frac{1}{2} < \mu < 1 < \lambda$, and let k be a symmetric bilinear form on $\mathbb{R}^4$. If $\|k-\eta_1\| \le \varepsilon < \min\{(1-\mu^2)/(1+\mu^2), (\lambda^2-1)/(\lambda^2+1)\}$ (operator norm), then k is Lorentzian with ∂_t timelike, and for $v=(1,u)$: $|u| \le \mu \Rightarrow k(v,v) < 0$, $|u| \ge \lambda \Rightarrow k(v,v) > 0$. The bound is preserved by $\chi k + (1-\chi)\eta_1$, $0 \le \chi \le 1$.*

Proof. $k(v,v) \le -1 + |u|^2 + \varepsilon(1+|u|^2)$ and $k(v,v) \ge (1-\varepsilon)|u|^2 - (1+\varepsilon)$. The spatial block is $\ge (1-\varepsilon)I > 0$ and $k(\partial_t,\partial_t) < 0$, which forces signature $(-,+,+,+)$. □

Constants and the auxiliary spacetime. In this order: fix μ,λ,ε as in Lemma 50; Lemma 49 gives ρ_0 and K; choose $r>0$ with $3r<\rho_0$, $6Kr \le \varepsilon$ and $8\lambda r < L$. Then $\overline{T_{2r}} \subset T_{3r}$; let χ be smooth, $\chi=1$ on a neighbourhood of $\overline{T_{2r}}$, $\operatorname{supp}\chi \subset T_{3r}$, and on $\mathbb{R}^4$ put $g' = \chi\,\Phi^*g + (1-\chi)\eta$. On T_{3r}, $\|\Phi^*g - \eta\| \le 6Kr \le \varepsilon$, so Lemma 50 applies to g' everywhere and to Φ^*g on T_{3r}. Every g'-causal vector therefore has $|d(s,y)/dt| \le \lambda$; t increases strictly along future causal curves; and an inextendible causal curve along which t had a finite limit would have convergent spatial coordinates and extend. Hence every level $\{t=c\}$ is a Cauchy hypersurface and $M' = (\mathbb{R}^4, g')$ is globally hyperbolic (MS Thm. 3.75), flat outside T_{3r}, and $\psi = \Phi^{-1} : \Phi(T_{2r}) \to T_{2r}$ is an isometry preserving orientation and time orientation.

Lemma 51 (The slab region is self-determined)**.** *Let $B_0 = \{s \in [-r/2, L+r/2],\ |y| \le r/2\}$, $\Sigma = \Phi(\{0\}\times B_0) \subset S$ and $\Sigma' = \{0\}\times B_0$. Then:*

(i) $D_M(\Sigma) \subseteq \Phi(T_{2r})$ and $D_{M'}(\Sigma') \subseteq T_{2r}$;

(ii) $\psi(D_M(\Sigma)) = D_{M'}(\Sigma')$;

(iii) $N = \operatorname{int} D_M(\Sigma)$ is causally convex in M, globally hyperbolic, connected, and contains x, b; likewise $\psi(N) = \operatorname{int} D_{M'}(\Sigma')$ in M'.

Proof. Escape paths. Let $q \in \partial T_{2r}$ with $0 < t(q) \le 2r$. If the projection (s_q, y_q) is not in B_0, follow the vertical segment down to $t=0$. Otherwise q lies on the lid $t=2r$ with $|y_q| \le r/2$; move with velocity $(dt, dy) = (-1, \frac{\mu}{2}\hat{y})$ for coordinate time $2r$, where $\hat{y}$ is a unit vector ($y_q/|y_q|$ if $y_q \neq 0$), ending at $|y| \in (r/2, 3r)$ because $\mu r > r/2$ and $r/2 + \mu r < 3r$. Either path lies in T_{3r}, has coordinate speed $< \mu$, so is timelike for Φ^*g and for g' (Lemma 50), and ends at a point of $\{t=0\}$ outside Σ', i.e. on $S \setminus \Sigma$ in M (Φ is injective). A past-directed causal curve starting on a Cauchy hypersurface never returns to it. The time-reversed paths serve points with $t(q) < 0$.

(i) Let $p \in D^+_M(\Sigma)$ with $p \notin \Phi(T_{2r})$, and let γ be a past-inextendible causal curve from p. It meets Σ; let σ_1 be its first point on Σ, and q the last point of $\partial\Phi(T_{2r})$ on the segment from p to σ_1. After q the segment lies in $\Phi(T_{2r})$, where t increases along future causal curves, so $t(q) > 0$. The segment $p \to q$, the escape path from q, and any past-inextendible continuation form a past-inextendible causal curve from p missing Σ, a contradiction. Time-reverse for D^-. The same argument works in M', where $\{t=0\}$ is a Cauchy hypersurface.

(ii) Let $p \in D^+_M(\Sigma)$, so $p \in \Phi(T_{2r})$ by (i), and let γ' be a past-inextendible g'-causal curve from $\psi(p)$. If $\psi(p) \in \Sigma'$ there is nothing to prove; otherwise $t(\psi(p)) > 0$. Along γ', $t \to -\infty$; let τ_0 be the first parameter at which $t=0$ and τ_{exit} the first at which γ' meets ∂T_{2r}. The chart and the agreement $g' = \Phi^*g$ extend to a neighbourhood of $\overline{T_{2r}}$, so segments of γ' up to either time pull

back by Φ. If $\tau_0 \le \tau_{\text{exit}}$ (equality included): were $\gamma'(\tau_0) \notin \Sigma'$, the pulled-back segment ends on $S \setminus \Sigma$ and, continued past S, is a past-inextendible M-curve from p missing Σ (points with $t > 0$ are not on S). If $\tau_{\text{exit}} < \tau_0$, then $t > 0$ at the exit point, and the pulled-back segment followed by the escape path in M again misses Σ. Both contradict $p \in D^+_M(\Sigma)$, so γ' meets Σ'. The reverse inclusion is the same argument with M and M' exchanged, using $D_{M'}(\Sigma') \subseteq T_{2r}$: no global coordinate t on M is needed, since a past-inextendible M-curve from a point of $\Phi(T_{2r})$ either leaves $\Phi(T_{2r})$ or reaches the Cauchy surface S, and the same two cases apply. D^- is time-reversed.

(iii) For $q \in D^+(\Sigma)$, $J^-(q) \cap J^+(\Sigma) \subseteq D^+(\Sigma)$: a past-inextendible curve from r in this set, prefixed by a causal curve $q \to r$, meets Σ; if only on the prefix, at z, then $\sigma \le r \le z$ for some $\sigma \in \Sigma$, and acausality gives $r = z \in \Sigma$. Hence $D(\Sigma)$ is causally convex: if $p, q \in D(\Sigma)$, r lies on a causal curve $p \to q$, and an inextendible causal curve λ through r misses Σ, then in the mixed case $p \in D^-$, $q \in D^+$ the curves $(p \to r) \cdot \lambda^+$ and $\lambda^- \cdot (r \to q)$ meet Σ on $p \to r$ and on $r \to q$, and acausality forces both meeting points to be r; the cases $p, q \in D^\pm$ follow from the inclusion just proved. For interior points choose $p^- \ll p$ and $q^+ \gg q$ in $\operatorname{int} D$; then $J(p,q) \subseteq I^+_{\text{s}}(p^-) \cap I^-_{\text{s}}(q^+) \subseteq D(\Sigma)$, an open set (MS Props. 2.16, 2.31), so N is causally convex, its causal diamonds are the compact diamonds of M, and N is globally hyperbolic (MS Def. 3.70). In M', $D_{\eta_\lambda}(\Sigma') \subseteq D_{g'}(\Sigma') \subseteq D_{\eta_\mu}(\Sigma')$ (narrower cones give larger domains of dependence). The first inclusion puts every relative-interior point of Σ', in particular x and b, in the interior; the second puts the projection of every interior point with $t \neq 0$ into $\operatorname{int} B_0$, so its vertical segment to $\{t = 0\}$ lies in $I^-(q^+) \cap I^+(z^-) \subseteq N'$ for suitable q^+, z^-, joining it inside N' to the connected set $\operatorname{int} \Sigma'$. Transfer by (ii). □

By Lemma 51, N is an object of GH and the inclusion $N \hookrightarrow M$ and $\psi|_N : N \to M'$ are morphisms. By (LC),

$$x \, B_M \, b \iff x \, B_N \, b \iff \psi(x) \, B_{M'} \, \psi(b). \tag{5}$$

Lemma 52 (Flat detour)**.** *Let P, Q lie in the flat region of M', be η_λ-spacelike, and be joined by a smooth embedded arc κ contained in a hyperplane $H = \{t = f(s)\}$ with f affine, $|f'| < 1/\lambda$, and at positive distance from $\overline{T_{3r}}$. Then P, Q are spacelike-separated in M', and there is a compact acausal slab $\Sigma'' \subset H$, containing P, Q in its relative interior, such that $N'' = \operatorname{int} D_{M'}(\Sigma'')$ lies in the flat region, equals $\operatorname{int} D_{\mathbb{M}}(\Sigma'')$ under the identity, and is an object of GH whose inclusions into M' and $\mathbb{M}$ are morphisms. Hence $P \, B_{M'} \, Q \iff P \, B_{\mathbb{M}} \, Q$.*

Proof. A g'-causal curve between P, Q would be η_λ-causal. Along g'-causal curves $\frac{d}{d\tau}(t - f(s)) \ge \dot{t}(1 - \lambda|f'|) > 0$ and $t - f(s) \to \pm\infty$, so H is a Cauchy hypersurface for g' and for η, crossed exactly once by every inextendible causal curve of either metric. Extend κ slightly past P and Q inside H, keeping it embedded and off $\overline{T_{3r}}$, and let Σ'' be its closed δ-neighbourhood in H. A boost along s followed by a time translation, an isometry of η preserving time orientation, makes H horizontal: in the new coordinates $(\tilde{t}, z)$, $\eta = -d\tilde{t}^2 + |dz|^2$ and $H = \{\tilde{t} = 0\}$. Put

$$W^\pm = \{(\tilde{t}, z) : \pm\tilde{t} \ge 0, \ \overline{B}(z, |\tilde{t}|) \subseteq \Sigma''\},$$

the two halves of $D_{\mathbb{M}}(\Sigma'')$. As $\delta \downarrow 0$ the sets Σ'' decrease to the extended arc, which is compact with empty interior in H. So the largest radius $h(\delta)$ of a closed ball in Σ'' tends to 0: a limit of balls of radius bounded below would lie in the arc. Take δ with $\delta + h(\delta)$ small compared with the $(\tilde{t}, z)$-distance from the extended arc to $\overline{T_{3r}}$. Every point of $W = W^+ \cup W^-$ has $z \in \Sigma''$ and $|\tilde{t}| \le h(\delta)$, so W is compact and has an open neighbourhood U on which $g' = \eta$. For p with $\tilde{t}(p) \ge 0$ let $C(p) = J^-_\eta(p) \cap \{\tilde{t} \ge 0\}$. If $p \in W^+$ then $C(p) \subseteq W^+$, because the balls are nested.

$W^+ \subseteq D^+_{M'}(\Sigma'')$: while a past-inextendible g'-causal curve from $p \in W^+$ stays in U and above H it is η-causal, so it lies in the compact set $C(p) \subset U$. Hence it reaches H inside $C(p) \cap H \subseteq \Sigma''$.

$D^+_{M'}(\Sigma'') \subseteq W^+$: suppose $p \in D^+_{M'}(\Sigma'') \setminus W^+$, and let γ be a past-inextendible g'-causal curve from p. It crosses H at a point of $\Sigma'' \subseteq W^+$. As W^+ is closed there is a last point of γ before which it lies outside W^+, so we can take q on γ above H, with $q \notin W^+$ and so close to W^+ that $C(q) \subset U$. Since $q \notin W^+$, a straight η-causal segment in $C(q)$ runs from q to a point of $H \setminus \Sigma''$, and it is g'-causal. Then γ up to q, this segment and any past-inextendible continuation form a past-inextendible g'-causal curve from p that crosses H only outside Σ'', a contradiction.

Time reversal handles W^-, so $D_{M'}(\Sigma'') = W = D_{\mathbb{M}}(\Sigma'')$. Its interior $N'' = \{|\tilde{t}| < \mathrm{dist}(z, H \setminus \Sigma'')\}$ lies in U, is connected and contains P, Q. The argument of Lemma 51(iii) shows that it is causally convex and globally hyperbolic in M' and in $\mathbb{M}$. □

Proof of Theorem 8. (a) Suppose $B_{\mathbb{M}} \neq \mathcal{U}$; by Theorem 7, $B_{\mathbb{M}} \subseteq J^-$ relates no spacelike pair. Let $x\, B_M\, b$ with x, b spacelike-separated, and transfer by (5). Take $w = (-4r, 0, 0, 0)$, $v = (4r, L, 0, 0)$ and the arc $t(s) = -4r + (8r/L)s$, $y(s) = (10r \sin(\pi s/L), 0)$, $0 \le s \le L$: where $|t| \le 3r$, $s/L \in [1/8, 7/8]$ and $|y| \ge 10r \sin(\pi/8) > 3r$, elsewhere $|t| > 3r$, and the slope $8r/L < 1/\lambda$. Vertical segments are η_μ-timelike, so $w \ll' \psi(x)$ and $\psi(b) \ll' v$; (CB) gives $w\, B'\, \psi(x)$ and $\psi(b)\, B'\, v$, and (T) gives $w\, B'\, v$. Lemma 52 gives $w\, B_{\mathbb{M}}\, v$ with w, v spacelike, a contradiction. If $x\, B_M\, b$ with $x \in J^+(b) \setminus \{b\}$, Lemma 47 gives $e \ll x$ spacelike to b, and $e\, B_M\, b$ by (CB) and (T), reducing to the previous case.

(b) Suppose $B_{\mathbb{M}} = \mathcal{U}$. For spacelike $x, b \in M$, transfer to M' and use $x' = (4r, 0, 0, 0)$, $b' = (-4r, L, 0, 0)$ with the arc $t(s) = 4r - (8r/L)s$, same $y(s)$: $\psi(x)\, B'\, x'$ and $b'\, B'\, \psi(b)$ by (CB), and $x'\, B'\, b'$ by Lemma 52, since $B_{\mathbb{M}} = \mathcal{U}$; so $x\, B_M\, b$. By Lemma 48 and (T), every pair is related. □

B Standpoint logic

Read $\Box A$ at b as "at every event $x \le b$, A" on the frame (M, R) with bRx iff $x \le b$. If A is a tenseless claim about what occurs at x, this is the reflexive past modality of Diodorean tense logic, and the standpoint reading of Fine and Lipman (Fine, 2005; Lipman, 2023) adds no formal content. R is reflexive and transitive, so S4 is valid, and in Minkowski spacetime the valid formulas are exactly those of S4.2 (Goldblatt, 1980). The axiom .2, $\Diamond\Box A \to \Box\Diamond A$, is valid at a world for all valuations iff any two of its successors have a common successor (Blackburn et al., 2001). So .2 is valid at b iff the subframe generated by b is directed, that is, iff any two events in $J^-(b)$ have a common causal past.

In a model conformal to $(\eta_-, \eta_+) \times \mathbb{R}^3$, two events have a common past iff $|X - Y| < \eta_x + \eta_y - 2\eta_-$, so if $\eta_- > -\infty$ the directedness fails in the past of every event and .2 is valid at no event. This is the particle-horizon structure of Rindler (Rindler, 1956) in modal form, and it holds equally with or without an inflationary era of finite conformal duration. The horizon problem of cosmology (Guth, 1981) is therefore not the failure of .2, which occurs with and without inflation, but the failure of particular instances: pairs of points on our last-scattering surface, at comoving separation up to $2(\eta_0 - \eta_{\mathrm{ls}})$, have no common past unless $\eta_0 - \eta_{\mathrm{ls}} < \eta_{\mathrm{ls}} - \eta_-$, which fails without inflation and holds with enough of it. Two cautions. If persistence is imposed on suitable tenseless past-event facts, rather than on centered claims such as "this event is present now," the corresponding valuations are constrained along the future direction, and the frame property that matters may concern common futures, which these models have when $\eta_+ = \infty$; the logic of persistent valuations is a separate question. And Lewis and van der Schaaf study causal modal logics of general spacetimes

(Lewis and van der Schaaf, 2026); we have not checked how their results bear on the past-directed frames used here. Whether S4 is exactly the logic of all past-directed globally hyperbolic frames we leave open.

C Status of the results

"Reviewed" means internal review with its record in the project repository, not refereeing. Every entry retains the conditions stated in the text.

Result	Status
Theorem 8 and corollaries	Reviewed, conditional on (LC), (T), (CB) and the off-shell premise. Lemma 52 reproved 31 October with one model review. Expert read pending
Theorems 14, 15, Proposition 16	Reviewed. A planned independent null-boundary audit did not finish; its curvature script, recovered 31 October, passes, but it left no report
Causal-set and energy-condition sections	Reviewed in the manuscript's own reviews. Separate audits were stopped unfinished
Theorem 24, models, Theorem 33	Reviewed; proofs checked. Priority search bounded. Overlap with Geroch, Manchak and Earman noted
Theorem 35	Reviewed research note; outline only here. Full proof: `research/horizons/mass-classification/RESULT.md` in the project repository. Existing lacuna results may imply it
Proposition 34	Reviewed. An unfinished referee's numerical control passes after two repairs to its script on 31 October
Theorem 37	Assembled from the cited results; rows checked by one model review
Proposition 36	Reviewed
Theorem 38	Finite-control audit accepted at bounded scope; regularity audited. Inherits external LOY inputs, including omitted proofs. Full proof: `research/lawful-completions/coordinator/` in the project repository. Priority comparison incomplete; much of the setting is in print
Lemma 40, Theorem 41, Proposition 39	Bounded audit of a specialization. External uniqueness and common-extension theorems used at selected read depth
Section 10	Proposition 42 elementary; horizon lemma and cylinder classification reviewed. Bigaj known through one worker's full-text report, printed pages not checked
Sections 11, 12	Coordinator arguments with bounded review; restricted counterinterpretation, not named theories
Section 13	New 31 October. Proposition 43 elementary. Albert, Price and Tooley read in part. Not yet red-teamed. Proposition 44 and Corollary 45 added 10 November: elementary, checked only by the session that drafted them; the corollary inherits Theorem 35 and the mirror. The deterministic case of reciprocity is Albert's (ch. 6, n. 11, read directly)
Proposition 10, Corollaries 11, 12	Added 10 November. Elementary; checked only by the session that drafted them. Prior-art search short; may be known in locally covariant field theory
Proposition 20, Remark 21	Added 10 November; checked only by the session that drafted them. An earlier project draft already noted that a curvature ordering is local. The converse stated in the remark rests on an unaudited note. The Fewster–Verch definitions were checked against arXiv v3 on 10 November
Proposition 30, Remark 29	Added 10 November; checked only by the session that

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